Base R: 120 Exercises with Solutions

Vectors · Operators · Functions · Data Frames · Lists · Matrices · Programming · Apply · Strings · Regex · Date/Time · I/O · Error Handling

Outline

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9. Date/Time, I/O, Error Handling, Advanced (Ex 111–120)

Vectors and Types (Ex 1–18)

Ex 1: Logical Indexing

Problem

Let `v <- c(14, -3, 22, 7, -8, 0, 31, 5, -1, 18)`.

- **(a)** Extract all elements strictly between 5 and 25
- **(b)** Replace every negative element with `0` in a copy `w`
- **(c)** How many elements are negative or exceed 20?
- **(d)** Find indices where $|v_i|$ exceeds its rank

Sol 1

```r
v <- c(14, -3, 22, 7, -8, 0, 31, 5, -1, 18)
v[v > 5 & v < 25]                    # (a)
```

```
# [1] 14 22  7 18
```

```r
w <- v; w[w < 0] <- 0; w             # (b)
```

```
#  [1] 14  0 22  7  0  0 31  5  0 18
```

```r
sum(v < 0 | v > 20)                  # (c)
```

```
# [1] 5
```

```r
which(abs(v) > rank(v))              # (d)
```

```
# [1]  1  2  3  4  5  7 10
```

Ex 2: Named Vectors

Problem

Create `stock <- c(AAPL=189.5, MSFT=415.2, GOOG=172.3, AMZN=185.7)`.

- **(a)** Access `GOOG` by name; add `NVDA = 875.4`
- **(b)** Compute percentage change from base 150; round to 1 dp
- **(c)** Return stocks above mean price, sorted descending
- **(d)** Rename all tickers to lowercase

Sol 2

```r
stock <- c(AAPL=189.5, MSFT=415.2, GOOG=172.3, AMZN=185.7)
stock["GOOG"]; stock["NVDA"] <- 875.4             # (a)
```

```
#  GOOG
# 172.3
```

```r
round((stock - 150) / 150 * 100, 1)               # (b)
```

```
#  AAPL  MSFT  GOOG  AMZN  NVDA
#  26.3 176.8  14.9  23.8 483.6
```

```r
sort(stock[stock > mean(stock)], decreasing=TRUE)  # (c)
```

```
#  NVDA  MSFT
# 875.4 415.2
```

```r
names(stock) <- tolower(names(stock)); names(stock)# (d)
```

```
# [1] "aapl" "msft" "goog" "amzn" "nvda"
```

Ex 3: `which`, `which.min`, `which.max`

Problem

Let `x <- c(3, 7, 2, 9, 1, 5, 8, 4, 6, 10)`.

- **(a)** Indices of elements > 6
- **(b)** Index of min and max
- **(c)** Second-largest value (no `sort`)
- **(d)** Local maxima: elements greater than both immediate neighbours (ignore endpoints)

Sol 3

```r
x <- c(3, 7, 2, 9, 1, 5, 8, 4, 6, 10)
which(x > 6)                                    # (a)
```

```
# [1]  2  4  7 10
```

```r
which.min(x); which.max(x)                      # (b)
```

```
# [1] 5
```

```
# [1] 10
```

```r
max(x[x != max(x)])                             # (c)
```

```
# [1] 9
```

```r
n <- length(x); idx <- 2:(n-1)                  # (d)
x[idx][x[idx] > x[idx-1] & x[idx] > x[idx+1]]
```

```
# [1] 7 9 8
```

Ex 4: Cumulative and Parallel Functions

Problem

Let `x <- c(2,5,3,8,1,7,4)` and `y <- c(6,2,9,1,4,3,8)`.

- **(a)** Running maximum of `x`
- **(b)** Running sum and product
- **(c)** Element-wise min and max
- **(d)** Verify $\sum \text{pmin}(x, y) + \sum \text{pmax}(x, y) = \sum x + \sum y$

Sol 4

```r
x <- c(2,5,3,8,1,7,4); y <- c(6,2,9,1,4,3,8)
cummax(x)                                          # (a)
```

```
# [1] 2 5 5 8 8 8 8
```

```r
cumsum(x); cumprod(x)                              # (b)
```

```
# [1]  2  7 10 18 19 26 30

# [1]    2   10   30  240  240 1680 6720
```

```r
pmin(x,y); pmax(x,y)                              # (c)
```

```
# [1] 2 2 3 1 1 3 4

# [1] 6 5 9 8 4 7 8
```

```r
sum(pmin(x,y)) + sum(pmax(x,y)) == sum(x) + sum(y)# (d)
```

```
# [1] TRUE
```

Ex 5: Set Operations

Problem

Let `a <- c(1,3,5,7,9,3,5)` and `b <- c(2,3,5,6,7,2)`.

- **(a)** Unique elements of `a`
- **(b)** Positions of duplicated values in `a`
- **(c)** $a \cup b$, $a \cap b$, $a \setminus b$
- **(d)** `match(c(3,5,4), b)` — explain the `NA`

Sol 5

```r
a <- c(1,3,5,7,9,3,5); b <- c(2,3,5,6,7,2)
unique(a)                                          # (a)
```

```
# [1] 1 3 5 7 9
```

```r
which(duplicated(a))                               # (b)
```

```
# [1] 6 7
```

```r
union(a,b); intersect(a,b); setdiff(a,b)          # (c)
```

```
# [1] 1 3 5 7 9 2 6

# [1] 3 5 7

# [1] 1 9
```

```r
match(c(3,5,4), b)  # 4 not found -> NA            # (d)
```

```
# [1]  2  3 NA
```

Ex 6: Recycling Rule

Problem

- **(a)** What does `c(1,2,3,4,5,6) + c(10,20)` return? Why?
- **(b)** Let `m <- matrix(1:12, nrow=3)`. What does `m + c(100,200,300)` do?
- **(c)** Extract odd-indexed and even-indexed elements of `1:12` using `c(TRUE,FALSE)`
- **(d)** Generate `1 0 1 0 1 0 1 0 1 0` using `rep`

Sol 6

```r
c(1,2,3,4,5,6) + c(10,20)                          # (a) recycled
```

```
# [1] 11 22 13 24 15 26
```

```r
m <- matrix(1:12, nrow=3)
m + c(100,200,300)    # adds per column (col-major) # (b)
```

```
#      [,1] [,2] [,3] [,4]
# [1,]  101  104  107  110
# [2,]  202  205  208  211
# [3,]  303  306  309  312
```

```r
x <- 1:12; x[c(T,F)]; x[c(F,T)]                  # (c)
```

```
# [1]  1  3  5  7  9 11

# [1]  2  4  6  8 10 12
```

```r
rep(c(1,0), 5)                                     # (d)
```

```
#  [1] 1 0 1 0 1 0 1 0 1 0
```

Ex 7: Sorting and Ranking

Problem

Let `x <- c(40,15,30,10,50,25)`.

- **(a)** Sort ascending and descending
- **(b)** Find `order(x)` and verify `x[order(x)] == sort(x)`
- **(c)** What rank does 30 receive?
- **(d)** Sort `c("banana","apple","cherry")` alphabetically

Sol 7

```r
x <- c(40,15,30,10,50,25)
sort(x); sort(x, decreasing=TRUE)                    # (a)
```

```
# [1] 10 15 25 30 40 50

# [1] 50 40 30 25 15 10
```

```r
ord <- order(x); all(x[ord] == sort(x))              # (b)
```

```
# [1] TRUE
```

```r
rank(x)  # 30 -> rank 4                              # (c)
```

```
# [1] 5 2 4 1 6 3
```

```r
sort(c("banana","apple","cherry"))                    # (d)
```

```
# [1] "apple"  "banana" "cherry"
```

Ex 8: NA, NULL, NaN, Inf

Problem

- **(a)** Given `x <- c(3, NA, -1, NaN, Inf, -Inf, 0)`, apply `is.na`, `is.nan`, `is.finite`, `is.infinite`. Why is `is.na(NaN)` TRUE?
- **(b)** Remove non-finite values and compute mean
- **(c)** Show `c(1, NULL, 3)` vs `c(1, NA, 3)` lengths
- **(d)** Compute `NA\textasciicircum{0}`, `NA | TRUE`, `NA \& FALSE`

Sol 8

```r
x <- c(3, NA, -1, NaN, Inf, -Inf, 0)
is.na(x); is.nan(x); is.finite(x)                    # (a)
```

```
# [1] FALSE  TRUE FALSE  TRUE FALSE FALSE FALSE

# [1] FALSE FALSE FALSE  TRUE FALSE FALSE FALSE

# [1]  TRUE FALSE  TRUE FALSE FALSE FALSE  TRUE
```

```r
mean(x[is.finite(x)])                                 # (b)
```

```
# [1] 0.6666667
```

```r
length(c(1, NULL, 3)); length(c(1, NA, 3))            # (c) 2 vs 3
```

```
# [1] 2

# [1] 3
```

```r
NA^0; NA | TRUE; NA & FALSE                           # (d)
```

```
# [1] 1

# [1] TRUE

# [1] FALSE
```

Ex 9: Type Coercion

Problem

- **(a)** What does `c(TRUE, 1L, 2.5, "x")` produce? What type?
- **(b)** Coerce `c("3","1.5","NA","7")` to numeric. What happens to `"NA"`?
- **(c)** Predict `as.integer(c(3.9, -2.7, TRUE, "5"))`
- **(d)** Why does `1 == "1"` return `TRUE`?

Sol 9

```r
x <- c(TRUE, 1L, 2.5, "x"); x; typeof(x)          # (a) character
```

```
# [1] "TRUE" "1"    "2.5"  "x"

# [1] "character"
```

```r
as.numeric(c("3","1.5","NA","7"))                  # (b) NA + warning
```

```
# [1] 3.0 1.5  NA 7.0
```

```r
as.integer(c(3.9, -2.7, TRUE, "5"))               # (c) 3,-2,1,5
```

```
# [1]  3 -2 NA  5
```

```r
1 == "1"  # 1 coerced to "1", then string compare  # (d)
```

```
# [1] TRUE
```

Ex 10: `seq` and `seq_along`

Problem

- **(a)** Generate `1.0 2.5 4.0 5.5 7.0 8.5 10.0` using `seq`
- **(b)** Show `1:length(x)` is unsafe when `x <- NULL`; fix with `seq\_along`
- **(c)** 6 equally-spaced values from 0 to 2π
- **(d)** What does `seq\_along(integer(0))` return?

Sol 10

```r
seq(1, 10, by=1.5)                                  # (a)
```

```
# [1]  1.0  2.5  4.0  5.5  7.0  8.5 10.0
```

```r
x <- NULL; 1:length(x)                              # (b) c(1,0) WRONG
```

```
# [1] 1 0
```

```r
seq_along(x)                                         # integer(0) SAFE
```

```
# integer(0)
```

```r
seq(0, 2*pi, length.out=6)                          # (c)
```

```
# [1] 0.000000 1.256637 2.513274 3.769911 5.026548 6.283185
```

```r
seq_along(integer(0))                                # (d) integer(0)
```

```
# integer(0)
```

Ex 11: `rep` Patterns

Problem

Using `rep`, generate:

- **(a)** `1 1 2 2 3 3 4 4 5 5`
- **(b)** `1 2 3 1 2 3 1 2 3 1 2 3`
- **(c)** `"A" "A" "A" "B" "B" "C"`
- **(d)** Alternating `TRUE FALSE` of length 20

Sol 11

```r
rep(1:5, each=2)                                    # (a)
```

```
#  [1] 1 1 2 2 3 3 4 4 5 5
```

```r
rep(1:3, times=4)                                   # (b)
```

```
#  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```r
rep(c("A","B","C"), c(3,2,1))                      # (c)
```

```
# [1] "A" "A" "A" "B" "B" "C"
```

```r
rep(c(TRUE,FALSE), 10)                              # (d)
```

```
#  [1]  TRUE FALSE  TRUE FALSE  TRUE FALSE  TRUE FALSE  TRUE FALSE  TRUE FALSE
# [13]  TRUE FALSE  TRUE FALSE  TRUE FALSE  TRUE FALSE
```

Ex 12: `%in%` and `match`

Problem

`haystack <- c(3,1,4,1,5,9,2,6,5,3)`; `needles <- c(5,7,2)`.

- **(a)** Which needles are present?
- **(b)** First occurrence position of each
- **(c)** Difference between `match("a", c("b","a","a"))` and `which(c("b","a","a") == "a")`
- **(d)** Filter `mtcars` rows where `cyl %in% c(4,6)`

Sol 12

```r
haystack <- c(3,1,4,1,5,9,2,6,5,3); needles <- c(5,7,2)
needles %in% haystack                                  # (a)
```

```
# [1]  TRUE FALSE  TRUE
```

```r
match(needles, haystack)                               # (b) NA for 7
```

```
# [1]  5 NA  7
```

```r
match("a", c("b","a","a"))                             # (c) 2 (first)
```

```
# [1] 2
```

```r
which(c("b","a","a") == "a")                           # c(2,3) (all)
```

```
# [1] 2 3
```

```r
nrow(mtcars[mtcars$cyl %in% c(4,6), ])                # (d)
```

```
# [1] 18
```

Ex 13: Vector Arithmetic

Problem

- **(a)** Compute $H_{100} = \sum_{k=1}^{100} \frac{1}{k}$ in one line
- **(b)** Compute $\sum_{k=0}^{50} \frac{(-1)^k}{k!}$. Compare with `exp(-1)`
- **(c)** For `x <- seq(0.1,2,by=0.1)`, compute $e^x \sin x - \ln x$, rounded to 4 dp
- **(d)** Compute `diff(cumsum(1:10))` and explain

Sol 13

```r
sum(1 / 1:100)                                    # (a) H_100
```

```
# [1] 5.187378
```

```r
k <- 0:50; sum((-1)^k / factorial(k)); exp(-1)    # (b)
```

```
# [1] 0.3678794
```

```
# [1] 0.3678794
```

```r
x <- seq(0.1, 2, by=0.1)
round(exp(x)*sin(x) - log(x), 4)                  # (c)
```

```
#  [1] 2.4129 1.8521 1.6029 1.4972 1.4836 1.5397 1.6540 1.8196 2.0320 2.2874
# [11] 2.5820 2.9122 3.2732 3.6597 4.0650 4.4809 4.8977 5.3036 5.6850 6.0257
```

```r
diff(cumsum(1:10))  # recovers 2:10                # (d)
```

```
# [1]  2  3  4  5  6  7  8  9 10
```

Ex 14: Attributes and `structure`

Problem

- **(a)** Create `a <- 1:4`; attach attributes `"unit"="kg"` and `"source"="lab"`. Print
- **(b)** Recreate using `structure()` in one call
- **(c)** `x <- 1:3` vs `y <- c(1,2,3)`: are they `identical`? `all.equal`? Why?
- **(d)** When to use `isTRUE(all.equal())` in an `if`?

Sol 14

```r
a <- 1:4; attr(a,"unit") <- "kg"; attr(a,"source") <- "lab"
attributes(a)                                          # (a)
```

```
# $unit
# [1] "kg"
#
# $source
# [1] "lab"
```

```r
b <- structure(1:4, unit="kg", source="lab")           # (b)
x <- 1:3; y <- c(1,2,3)
identical(x, y); all.equal(x, y)                       # (c) F; T
```

```
# [1] FALSE
```

```
# [1] TRUE
```

```r
isTRUE(all.equal(0.1+0.2, 0.3))                       # (d) TRUE
```

```
# [1] TRUE
```

Ex 15: Factors

Problem

- **(a)** Create factor `f <- factor(c("low","med","high","med","low"))`. What are the levels?
- **(b)** Create an ordered factor with levels `low < med < high`. Test `f > "low"`
- **(c)** Use `table(f)` to count occurrences
- **(d)** What does `as.integer(f)` return and why?

Sol 15

```r
f <- factor(c("low","med","high","med","low"))
levels(f)                                          # (a) alphabetical
```

```
# [1] "high" "low"  "med"
```

```r
fo <- factor(f, levels=c("low","med","high"), ordered=TRUE)
fo > "low"                                         # (b)
```

```
# [1] FALSE  TRUE  TRUE  TRUE FALSE
```

```r
table(f)                                           # (c)
```

```
# f
# high  low  med
#    1    2    2
```

```r
as.integer(f)  # underlying integer codes          # (d)
```

```
# [1] 2 3 1 3 2
```

Ex 16: Floating Point

Problem

- **(a)** Is `0.1 + 0.2 == 0.3` TRUE or FALSE? Why?
- **(b)** Use `all.equal` to compare correctly
- **(c)** What is `trunc(-3.7)` vs `floor(-3.7)`?
- **(d)** Demonstrate banker's rounding: `round(0.5)`, `round(1.5)`, `round(2.5)`

Sol 16

```r
0.1 + 0.2 == 0.3  # FALSE (floating point)        # (a)
```

```
# [1] FALSE
```

```r
all.equal(0.1 + 0.2, 0.3)                         # (b) TRUE
```

```
# [1] TRUE
```

```r
trunc(-3.7); floor(-3.7)  # -3 vs -4              # (c)
```

```
# [1] -3

# [1] -4
```

```r
round(0.5); round(1.5); round(2.5)  # 0; 2; 2    # (d)
```

```
# [1] 0

# [1] 2

# [1] 2
```

Ex 17: Appending and Inserting

Problem

- **(a)** Start with `v <- 1:5`. Append `c(10,11)` to the end
- **(b)** Insert 99 after position 3 using `append`
- **(c)** What happens if you assign `v[10] <- 42` when `length(v)` is 7?
- **(d)** Remove the 3rd element without using negative indexing (use logical)

Sol 17

```r
v <- 1:5; v <- c(v, c(10,11)); v                    # (a)
```

```
# [1]  1  2  3  4  5 10 11
```

```r
append(v, 99, after=3)                               # (b)
```

```
# [1]  1  2  3 99  4  5 10 11
```

```r
v[10] <- 42; v   # NAs fill gaps                     # (c)
```

```
#  [1]  1  2  3  4  5 10 11 NA NA 42
```

```r
v[seq_along(v) != 3]                                 # (d)
```

```
# [1]  1  2  4  5 10 11 NA NA 42
```

Ex 18: `sample` and `sequence`

Problem

- **(a)** Simulate 20 dice rolls with `sample`
- **(b)** Use `sample` to shuffle `letters[1:10]`
- **(c)** Using `sequence`, generate `1 2 3 4  1 2 3  1 2  1`
- **(d)** Generate `5 4 3 2 1  6 5 4 3 2 1` using `sequence`

Sol 18

```r
set.seed(42)
sample(1:6, 20, replace=TRUE)                    # (a)
```

```
#  [1] 1 5 1 1 2 4 2 2 1 4 1 5 6 4 2 2 3 1 1 3
```

```r
sample(letters[1:10])                             # (b)
```

```
#  [1] "d" "e" "g" "i" "h" "j" "b" "c" "f" "a"
```

```r
sequence(4:1)                                     # (c)
```

```
#  [1] 1 2 3 4 1 2 3 1 2 1
```

```r
sequence(5:6, from=5:6, by=-1)                    # (d)
```

```
#  [1] 5 4 3 2 1 6 5 4 3 2 1
```

Operators and Common Functions (Ex 19–26)

Ex 19: Operator Precedence

Problem

- **(a)** Without running, predict: `2 + 3 * 4`, `2\textasciicircum{3^2}`, `1:3 + 1`, `1:(3+1)`
- **(b)** Why does `-2\textasciicircum{2}` give `-4` instead of 4?
- **(c)** Predict: `TRUE | FALSE \& FALSE`
- **(d)** Write `x <- 5; x %% 3 == 2 \& x > 3` and trace operator precedence

Sol 19

```r
2 + 3 * 4     # 14: * before +                    # (a)
```

```
# [1] 14
```

```r
2^3^2          # 512: ^ right-associative = 2^9
```

```
# [1] 512
```

```r
1:3 + 1        # 2 3 4: : before +
```

```
# [1] 2 3 4
```

```r
1:(3+1)        # 1 2 3 4
```

```
# [1] 1 2 3 4
```

```r
-2^2           # -4: ^ before unary -              # (b)
```

```
# [1] -4
```

```r
TRUE | FALSE & FALSE  # TRUE: & before |           # (c)
```

```
# [1] TRUE
```

```r
x <- 5; x %% 3 == 2 & x > 3  # TRUE               # (d)
```

```
# [1] TRUE
```

Ex 20: Math Functions

Problem

- **(a)** Compute `floor`, `ceiling`, `trunc`, `round` of `-5.7`
- **(b)** Compute $\binom{10}{3}$ and verify with `factorial`
- **(c)** Compute `119 %/% 13` and `119 %% 13`. Verify `119 == 13 * (119 %/% 13) + 119 %% 13`
- **(d)** `signif(12345678, 4)`

Sol 20

```r
floor(-5.7); ceiling(-5.7); trunc(-5.7); round(-5.7) # (a)
```

```
# [1] -6

# [1] -5

# [1] -5

# [1] -6
```

```r
choose(10, 3); factorial(10)/(factorial(3)*factorial(7)) # (b)
```

```
# [1] 120

# [1] 120
```

```r
119 %/% 13; 119 %% 13                                    # (c)
```

```
# [1] 9

# [1] 2
```

```r
119 == 13 * (119 %/% 13) + 119 %% 13                     # TRUE
```

```
# [1] TRUE
```

```r
signif(12345678, 4)                                       # (d)
```

Ex 21: Statistical Summary

Problem

`v <- c(4, 7, 2, 9, 1, 5, 8, 3, 6, 10)`.

- **(a)** mean, median, var, sd, range
- **(b)** `quantile(v, c(0.25, 0.75))` and IQR
- **(c)** Compute z-scores: $(v - \bar{v})/s$. How many exceed $|z| > 1$?
- **(d)** Use `diff(v)` to find the largest jump

Sol 21

```r
v <- c(4, 7, 2, 9, 1, 5, 8, 3, 6, 10)
mean(v); median(v); var(v); sd(v); range(v)        # (a)
```

```
# [1] 5.5

# [1] 5.5

# [1] 9.166667

# [1] 3.02765

# [1]  1 10
```

```r
quantile(v, c(0.25, 0.75)); IQR(v)                # (b)
```

```
#  25%  75%
# 3.25 7.75

# [1] 4.5
```

```r
z <- (v - mean(v)) / sd(v); sum(abs(z) > 1)       # (c)
```

```
# [1] 4
```

```r
max(abs(diff(v)))                                  # (d) largest jump
```

```
# [1] 8
```

Ex 22: `table` and `prop.table`

Problem

`x <- c("A","B","A","C","B","A","C","C","B","A")`.

- **(a)** Frequency table of `x`
- **(b)** Proportion table
- **(c)** Create `y <- c(1,1,2,1,2,2,1,2,1,2)`. Cross-tabulate `x` and `y`; add margins
- **(d)** Which category of `x` is most frequent?

Sol 22

```r
x <- c("A","B","A","C","B","A","C","C","B","A")
table(x)                                                    # (a)
```

```
# x
# A B C
# 4 3 3
```

```r
prop.table(table(x))                                        # (b)
```

```
# x
#   A   B   C
# 0.4 0.3 0.3
```

```r
y <- c(1,1,2,1,2,2,1,2,1,2)
addmargins(table(x, y))                                     # (c)
```

```
#       y
# x       1  2 Sum
#   A     1  3   4
#   B     2  1   3
#   C     2  1   3
#   Sum   5  5  10
```

```r
names(which.max(table(x)))                                  # (d) "A"
```

```
# [1] "A"
```

Ex 23: Harmonic Series Convergence

Problem

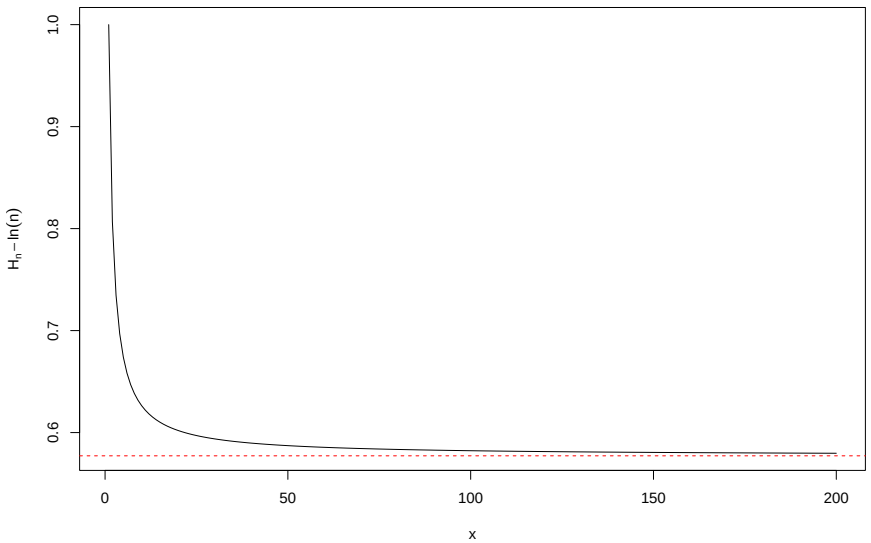
- **(a)** Compute $H_n = \sum_{k=1}^{n} \frac{1}{k}$ for $n \in \{10, 100, 1000, 10000\}$
- **(b)** Compare each H_n with $\ln(n) + \gamma$ where $\gamma \approx 0.5772$
- **(c)** Plot $H_n - \ln n$ for $n = 1, \ldots, 200$

Sol 23

```r
ns <- c(10, 100, 1000, 10000)                          # (a)
Hn <- sapply(ns, function(n) sum(1/1:n))
data.frame(n=ns, Hn=round(Hn,4),
           approx=round(log(ns)+0.5772,4))             # (b)
```

```
#       n     Hn approx
# 1    10 2.9290 2.8798
# 2   100 5.1874 5.1824
# 3  1000 7.4855 7.4850
# 4 10000 9.7876 9.7875
```

```r
x <- 1:200; y <- cumsum(1/x) - log(x)                  # (c)
plot(x, y, type="l", ylab=expression(H[n]-ln(n)))
abline(h=0.5772, col="red", lty=2)
```



Ex 24: `pmin`/`pmax` Application

Problem

- **(a)** Clip vector `v <- c(-5,3,12,-2,8,25)` to range $[0, 10]$ using `pmin` and `pmax`
- **(b)** Given two portfolios `a <- c(5,8,3,9)` and `b <- c(7,2,6,4)`, compute the "best of both" portfolio
- **(c)** Compute $\sum_{k=1}^{100} \min(2^k, k^4)$ without loops

Sol 24

```r
v <- c(-5, 3, 12, -2, 8, 25)
pmin(pmax(v, 0), 10)                                  # (a) clip to [0,10]
```

```
# [1]  0  3 10  0  8 10
```

```r
a <- c(5,8,3,9); b <- c(7,2,6,4)
pmax(a, b)                                            # (b)
```

```
# [1] 7 8 6 9
```

```r
k <- 1:100; sum(pmin(2^k, k^4))                      # (c)
```

```
# [1] 2050220551
```

Ex 25: Cumulative Functions

Problem

`x <- c(3, -1, 4, -1, 5, -9, 2, 6)`.

- **(a)** Compute `cumsum(x)`. At which index does it first become negative?
- **(b)** Maximum drawdown: $\max_i(\text{cummax}_i - \text{cumsum}_i)$
- **(c)** Compute `cumprod(1 + c(0.05, -0.02, 0.03, -0.01))` (portfolio growth)

Sol 25

```r
x <- c(3, -1, 4, -1, 5, -9, 2, 6)
cs <- cumsum(x); cs                                  # (a)
```

```
# [1]  3  2  6  5 10  1  3  9
```

```r
which(cs < 0)[1]                                     # first negative
```

```
# [1] NA
```

```r
cm <- cummax(cs); max(cm - cs)                       # (b) max drawdown
```

```
# [1] 9
```

```r
cumprod(1 + c(0.05, -0.02, 0.03, -0.01))            # (c) growth
```

```
# [1] 1.050000 1.029000 1.059870 1.049271
```

Ex 26: `order` and Data Sorting

Problem

```
d <- data.frame(name=c("Zara","Amy","Mike","Bob"), score=c(88,95,72,95),
age=c(22,20,25,21)).
```

- **(a)** Sort by score ascending
- **(b)** Sort by score descending, ties broken by age ascending
- **(c)** Find the name of the person with the highest score (youngest if tied)

Sol 26

```r
d <- data.frame(name=c("Zara","Amy","Mike","Bob"),
                score=c(88,95,72,95), age=c(22,20,25,21))
d[order(d$score), ]                                  # (a)
```

```
#   name score age
# 3 Mike    72  25
# 1 Zara    88  22
# 2  Amy    95  20
# 4  Bob    95  21
```

```r
d[order(-d$score, d$age), ]                          # (b)
```

```
#   name score age
# 2  Amy    95  20
# 4  Bob    95  21
# 1 Zara    88  22
# 3 Mike    72  25
```

```r
d[order(-d$score, d$age), ]$name[1]                  # (c) Bob
```

```
# [1] "Amy"
```

Data Frames (Ex 27–40)

Ex 27: Creating and Inspecting

Problem

- **(a)** Create a data frame with columns `name` (Alice, Bob, Cathy), `age` (25, 30, 28), `score` (88.5, 92.3, 76.8)
- **(b)** Show `str`, `dim`, `names`, `summary`
- **(c)** What class is `df\$name`?
- **(d)** Access `score` of row 2 by name and by position

Sol 27

```r
df <- data.frame(name=c("Alice","Bob","Cathy"),
                 age=c(25,30,28), score=c(88.5,92.3,76.8))
str(df); dim(df); names(df)                          # (b)
```

```
# 'data.frame': 3 obs. of  3 variables:
#  $ name : chr  "Alice" "Bob" "Cathy"
#  $ age  : num  25 30 28
#  $ score: num  88.5 92.3 76.8

# [1] 3 3

# [1] "name"  "age"   "score"
```

```r
class(df$name)                                       # (c)
```

```
# [1] "character"
```

```r
df$score[2]; df[2, "score"]                          # (d)
```

```
# [1] 92.3

# [1] 92.3
```

Ex 28: Row Filtering

Problem

Using `tg <- ToothGrowth`:

- **(a)** Rows where `supp == "OJ"` and `dose >= 1`
- **(b)** Mean tooth length for VC vs OJ
- **(c)** Rows where `len` is above its overall median
- **(d)** Same as (a) but using `subset()`

Sol 28

```r
tg <- ToothGrowth
head(tg[tg$supp == "OJ" & tg$dose >= 1, ], 4)     # (a)
```

```
#     len supp dose
# 41 19.7   OJ    1
# 42 23.3   OJ    1
# 43 23.6   OJ    1
# 44 26.4   OJ    1
```

```r
tapply(tg$len, tg$supp, mean)                     # (b)
```

```
#       OJ       VC
# 20.66333 16.96333
```

```r
head(tg[tg$len > median(tg$len), ], 4)            # (c)
```

```
#     len supp dose
# 15 22.5   VC    1
# 21 23.6   VC    2
# 23 33.9   VC    2
# 24 25.5   VC    2
```

```r
head(subset(tg, supp=="OJ" & dose >= 1), 4)       # (d)
```

```
#     len supp dose
# 41 19.7   OJ    1
# 42 23.3   OJ    1
```

Ex 29: Adding and Removing Columns

Problem

Start with `df <- data.frame(x=1:5, y=c(10,20,30,40,50))`.

- **(a)** Add column `z = x * y`
- **(b)** Add column `label` using `paste0("item", 1:5)`
- **(c)** Remove column `z`
- **(d)** Rename `y` to `value`

Sol 29

```r
df <- data.frame(x=1:5, y=c(10,20,30,40,50))
df$z <- df$x * df$y                                  # (a)
df$label <- paste0("item", 1:5)                       # (b)
df$z <- NULL                                          # (c)
names(df)[names(df)=="y"] <- "value"; df              # (d)
```

```
#   x value label
# 1 1    10 item1
# 2 2    20 item2
# 3 3    30 item3
# 4 4    40 item4
# 5 5    50 item5
```

Ex 30: `with()` and `within()`

Problem

Using `tg <- ToothGrowth`:

- **(a)** Use `with()` to compute `mean(len[dose == 2])`
- **(b)** Use `within()` to add `log\_len <- log(len)` and `len\_centered <- len - mean(len)`

Sol 30

```r
tg <- ToothGrowth
with(tg, mean(len[dose == 2]))                          # (a)
```

```
# [1] 26.1
```

```r
tg2 <- within(tg, {
  log_len <- log(len)
  len_centered <- len - mean(len)
})
head(tg2, 3)                                            # (b)
```

```
#    len supp dose len_centered  log_len
# 1  4.2   VC  0.5   -14.613333 1.435085
# 2 11.5   VC  0.5    -7.313333 2.442347
# 3  7.3   VC  0.5   -11.513333 1.987874
```

Ex 31: Handling NAs

Problem

`df <- data.frame(x=c(1,NA,3,4,NA), y=c(NA,2,3,NA,5), z=c(1,2,3,4,5)).`

- **(a)** Which rows are complete?
- **(b)** Remove all incomplete rows
- **(c)** Replace NAs in column `x` with the mean of `x` (ignoring NAs)
- **(d)** Count NAs per column

Sol 31

```r
df <- data.frame(x=c(1,NA,3,4,NA), y=c(NA,2,3,NA,5), z=1:5)
complete.cases(df)                                    # (a)
```

```
# [1] FALSE FALSE  TRUE FALSE FALSE
```

```r
df[complete.cases(df), ]                              # (b)
```

```
#   x y z
# 3 3 3 3
```

```r
df$x[is.na(df$x)] <- mean(df$x, na.rm=TRUE); df   # (c)
```

```
#          x  y z
# 1 1.000000 NA 1
# 2 2.666667  2 2
# 3 3.000000  3 3
# 4 4.000000 NA 4
# 5 2.666667  5 5
```

```r
colSums(is.na(df))                                    # (d)
```

```
# x y z
# 0 2 0
```

Ex 32: `merge()`

Problem

```
students <- data.frame(id=1:4, name=c("A","B","C","D")). grades <-
data.frame(sid=c(1,2,2,4), course=c("Math","Math","Eng","Eng"),
grade=c(85,90,78,92)).
```

- **(a)** Inner join on id
- **(b)** Left join (keep all students)
- **(c)** How many rows in each result? Why?

Sol 32

```r
students <- data.frame(id=1:4, name=c("A","B","C","D"))
grades <- data.frame(sid=c(1,2,2,4),
  course=c("Math","Math","Eng","Eng"), grade=c(85,90,78,92))
merge(students, grades, by.x="id", by.y="sid")     # (a) inner
```

```
#   id name course grade
# 1  1    A   Math    85
# 2  2    B   Math    90
# 3  2    B    Eng    78
# 4  4    D    Eng    92
```

```r
merge(students, grades, by.x="id", by.y="sid",
      all.x=TRUE)                                  # (b) left
```

```
#   id name course grade
# 1  1    A   Math    85
# 2  2    B   Math    90
# 3  2    B    Eng    78
# 4  3    C   <NA>    NA
# 5  4    D    Eng    92
```

```r
# inner: 4 rows; left: 5 rows (C has NA)           # (c)
```

Ex 33: `aggregate()`

Problem

Using `mtcars`:

- **(a)** Mean `mpg` by `cyl`
- **(b)** Mean and sd of `mpg` by `cyl` and `am`
- **(c)** Max `hp` by `cyl`

Sol 33

```r
aggregate(mpg ~ cyl, data=mtcars, FUN=mean)        # (a)
```

```
#   cyl      mpg
# 1   4 26.66364
# 2   6 19.74286
# 3   8 15.10000
```

```r
aggregate(mpg ~ cyl + am, data=mtcars,
  FUN=function(x) c(m=mean(x), s=sd(x)))           # (b)
```

```
#   cyl am      mpg.m      mpg.s
# 1   4  0 22.9000000  1.4525839
# 2   6  0 19.1250000  1.6317169
# 3   8  0 15.0500000  2.7743959
# 4   4  1 28.0750000  4.4838599
# 5   6  1 20.5666667  0.7505553
# 6   8  1 15.4000000  0.5656854
```

```r
aggregate(hp ~ cyl, data=mtcars, FUN=max)          # (c)
```

```
#   cyl  hp
# 1   4 113
# 2   6 175
# 3   8 335
```

Ex 34: `cbind` and `rbind`

Problem

- **(a)** Create two 1-column data frames and `cbind` them
- **(b)** Create two data frames with same columns and `rbind`
- **(c)** Use `do.call(rbind, list\_of\_dfs)` with 3 data frames

Sol 34

```r
d1 <- data.frame(a=1:3); d2 <- data.frame(b=4:6)
cbind(d1, d2)                                          # (a)
```

```
#   a b
# 1 1 4
# 2 2 5
# 3 3 6
```

```r
e1 <- data.frame(x="a", y=1); e2 <- data.frame(x="b", y=2)
rbind(e1, e2)                                          # (b)
```

```
#   x y
# 1 a 1
# 2 b 2
```

```r
dfs <- list(data.frame(x=1,y=2), data.frame(x=3,y=4),
            data.frame(x=5,y=6))
do.call(rbind, dfs)                                    # (c)
```

```
#   x y
# 1 1 2
# 2 3 4
# 3 5 6
```

Ex 35: Data Frame as Named List

Problem

- **(a)** Show `is.list(mtcars)` is TRUE. What does `typeof(mtcars)` return?
- **(b)** Access the `mpg` column using `[[` notation
- **(c)** What is the difference between `mtcars["mpg"]` and `mtcars[["mpg"]]`?
- **(d)** Convert `mtcars[1:3, 1:3]` to a list and back

Sol 35

```r
is.list(mtcars); typeof(mtcars)                    # (a) TRUE; "list"
```

```
# [1] TRUE

# [1] "list"
```

```r
head(mtcars[["mpg"]])                              # (b)
```

```
# [1] 21.0 21.0 22.8 21.4 18.7 18.1
```

```r
class(mtcars["mpg"]); class(mtcars[["mpg"]])       # (c) df vs numeric
```

```
# [1] "data.frame"

# [1] "numeric"
```

```r
l <- as.list(mtcars[1:3, 1:3]); as.data.frame(l)  # (d)
```

```
#    mpg cyl disp
# 1 21.0   6  160
# 2 21.0   6  160
# 3 22.8   4  108
```

Ex 36: `split()` on Data Frames

Problem

- **(a)** Split `iris` by `Species`. What type is the result?
- **(b)** Compute mean `Sepal.Length` per species using `split` + `sapply`
- **(c)** Recombine with `do.call(rbind, ...)`

Sol 36

```r
sp <- split(iris, iris$Species)
class(sp); names(sp)                                  # (a) list
```

```
# [1] "list"

# [1] "setosa"     "versicolor" "virginica"
```

```r
sapply(sp, function(d) mean(d$Sepal.Length))          # (b)
```

```
#     setosa versicolor  virginica
#      5.006      5.936      6.588
```

```r
head(do.call(rbind, sp), 4)                           # (c)
```

```
#          Sepal.Length Sepal.Width Petal.Length Petal.Width Species
# setosa.1          5.1         3.5          1.4         0.2  setosa
# setosa.2          4.9         3.0          1.4         0.2  setosa
# setosa.3          4.7         3.2          1.3         0.2  setosa
# setosa.4          4.6         3.1          1.5         0.2  setosa
```

Ex 37: `subset()` with `select`

Problem

Using `mtcars`:

- **(a)** Select only `mpg`, `cyl`, `hp` for cars with `mpg > 25`
- **(b)** Exclude `drat` and `wt` columns
- **(c)** Why can you write `subset(mtcars, mpg > 25)` without quoting `mpg`?

Sol 37

```r
subset(mtcars, mpg > 25, select=c(mpg, cyl, hp))  # (a)
```

```
#                  mpg cyl  hp
# Fiat 128        32.4   4  66
# Honda Civic     30.4   4  52
# Toyota Corolla 33.9   4  65
# Fiat X1-9       27.3   4  66
# Porsche 914-2  26.0   4  91
# Lotus Europa    30.4   4 113
```

```r
head(subset(mtcars, select=-c(drat, wt)), 3)       # (b)
```

```
#                 mpg cyl disp  hp  qsec vs am gear carb
# Mazda RX4      21.0   6  160 110 16.46  0  1    4    4
# Mazda RX4 Wag 21.0   6  160 110 17.02  0  1    4    4
# Datsun 710     22.8   4  108  93 18.61  1  1    4    1
```

```r
# (c) subset uses non-standard evaluation: it captures
# the expression and evaluates it within the data frame
```

Ex 38: Sorting with Multiple Keys

Problem

```
df <- data.frame(dept=c("A","B","A","B","A"),
name=c("Zoe","Amy","Bob","Cal","Amy"), salary=c(50,60,70,55,65)).
```

- **(a)** Sort by dept ascending, then salary descending
- **(b)** Find the highest-paid person in each department (using `order` + `!duplicated`)

Sol 38

```r
df <- data.frame(dept=c("A","B","A","B","A"),
  name=c("Zoe","Amy","Bob","Cal","Amy"), salary=c(50,60,70,55,65))
df[order(df$dept, -df$salary), ]                    # (a)
```

```
#   dept name salary
# 3    A  Bob     70
# 5    A  Amy     65
# 1    A  Zoe     50
# 2    B  Amy     60
# 4    B  Cal     55
```

```r
sorted <- df[order(df$dept, -df$salary), ]
sorted[!duplicated(sorted$dept), ]                  # (b)
```

```
#   dept name salary
# 3    A  Bob     70
# 2    B  Amy     60
```

Ex 39: Column-wise Operations

Problem

- **(a)** Use `colMeans` on `mtcars[,1:4]`
- **(b)** Use `sapply` to find `max` of each column
- **(c)** Standardize (z-score) all numeric columns using `scale()`. Compare with manual `sweep`

Sol 39

```r
colMeans(mtcars[,1:4])                                  # (a)
```

```
#       mpg       cyl      disp        hp
#  20.09062   6.18750 230.72188 146.68750
```

```r
sapply(mtcars[,1:4], max)                               # (b)
```

```
#   mpg   cyl  disp    hp
#  33.9   8.0 472.0 335.0
```

```r
head(scale(mtcars[,1:2]), 3)                            # (c)
```

```
#                     mpg        cyl
# Mazda RX4     0.1508848 -0.1049878
# Mazda RX4 Wag 0.1508848 -0.1049878
# Datsun 710    0.4495434 -1.2248578
```

```r
m <- as.matrix(mtcars[,1:2])
head(sweep(sweep(m, 2, colMeans(m)), 2, apply(m,2,sd), "/"), 3)
```

```
#                     mpg        cyl
# Mazda RX4     0.1508848 -0.1049878
# Mazda RX4 Wag 0.1508848 -0.1049878
# Datsun 710    0.4495434 -1.2248578
```

Ex 40: Building a Gradebook

Problem

- **(a)** Create `grades <- data.frame(name=c("A","B","C","D","E"), hw=c(85,92,78,95,88), exam=c(90,85,92,88,76))`
- **(b)** Add `final = 0.4*hw + 0.6*exam`
- **(c)** Add `letter` using nested `ifelse` (A/B/C/F)
- **(d)** Sort by `final` descending

Sol 40

```r
grades <- data.frame(name=c("A","B","C","D","E"),
  hw=c(85,92,78,95,88), exam=c(90,85,92,88,76))
grades$final <- 0.4*grades$hw + 0.6*grades$exam   # (b)
grades$letter <- ifelse(grades$final>=90,"A",      # (c)
  ifelse(grades$final>=80,"B",
  ifelse(grades$final>=70,"C","F")))
grades[order(-grades$final), ]                     # (d)
```

```
#   name hw exam final letter
# 4    D 95   88  90.8      A
# 1    A 85   90  88.0      B
# 2    B 92   85  87.8      B
# 3    C 78   92  86.4      B
# 5    E 88   76  80.8      B
```

Lists (Ex 41–52)

Ex 41: Creating and Accessing Lists

Problem

- **(a)** Create `l <- list(name="Alice", scores=c(85,92,78), passed=TRUE)`
- **(b)** Access `scores` with `\$`, `[[]]`, and `[]`. Show the different classes
- **(c)** Add element `grade = "B"`. Remove `passed`
- **(d)** What is `length(l)` before and after?

Sol 41

```r
l <- list(name="Alice", scores=c(85,92,78), passed=TRUE)
l$scores; l[["scores"]]; l["scores"]                # (b)
```

```
# [1] 85 92 78

# [1] 85 92 78

# $scores
# [1] 85 92 78
```

```r
class(l$scores); class(l["scores"])                 # vector vs list
```

```
# [1] "numeric"

# [1] "list"
```

```r
l$grade <- "B"; l$passed <- NULL                    # (c)
length(l)  # now 3                                  # (d)
```

```
# [1] 3
```

Ex 42: `unlist()` Behavior

Problem

- **(a)** `l <- list(a=1:3, b=4:5)`. What does `unlist(l)` produce? Explain the names
- **(b)** `unlist(l, use.names=FALSE)`
- **(c)** `l2 <- list(x="hello", y=1:3)`. What type is `unlist(l2)`? Why?
- **(d)** Mean of a list of numbers: `list(10, 20, 30)`

Sol 42

```r
l <- list(a=1:3, b=4:5)
unlist(l)                # named: a1 a2 a3 b1 b2   # (a)
```

```
# a1 a2 a3 b1 b2
#  1  2  3  4  5
```

```r
unlist(l, use.names=FALSE)                          # (b)
```

```
# [1] 1 2 3 4 5
```

```r
unlist(list(x="hello", y=1:3))  # all character    # (c)
```

```
#       x      y1      y2      y3
# "hello"     "1"     "2"     "3"
```

```r
mean(unlist(list(10, 20, 30)))                      # (d) 20
```

```
# [1] 20
```

Ex 43: Nested Lists

Problem

- **(a)** Create `student <- list(info=list(name="Bob", age=20), grades=list(math=90, eng=85))`
- **(b)** Access Bob's math grade
- **(c)** Add `grades\$sci = 92`
- **(d)** Use `str()` to visualize the structure

Sol 43

```r
student <- list(info=list(name="Bob", age=20),
                grades=list(math=90, eng=85))
student$grades$math                                  # (b) 90
```

```
# [1] 90
```

```r
student$grades$sci <- 92                             # (c)
str(student)                                         # (d)
```

```
# List of 2
#  $ info  :List of 2
#   ..$ name: chr "Bob"
#   ..$ age : num 20
#  $ grades:List of 3
#   ..$ math: num 90
#   ..$ eng : num 85
#   ..$ sci : num 92
```

Ex 44: `split()` and Recombine

Problem

- **(a)** `split(mtcars\$mpg, mtcars\$cyl)` — what type?
- **(b)** Compute `mean` per group using `sapply`
- **(c)** Same using `tapply` in one line
- **(d)** `split(1:10, rep(c("odd","even"), 5))`

Sol 44

```r
sp <- split(mtcars$mpg, mtcars$cyl); class(sp)    # (a) list
```

```
# [1] "list"
```

```r
sapply(sp, mean)                                   # (b)
```

```
#        4        6        8
# 26.66364 19.74286 15.10000
```

```r
tapply(mtcars$mpg, mtcars$cyl, mean)               # (c)
```

```
#        4        6        8
# 26.66364 19.74286 15.10000
```

```r
split(1:10, rep(c("odd","even"), 5))               # (d)
```

```
# $even
# [1]  2  4  6  8 10
#
# $odd
# [1] 1 3 5 7 9
```

Ex 45: List Concatenation

Problem

- **(a)** Concatenate `list(a=1)` and `list(b=2)` with `c()`
- **(b)** What happens with `c(list(a=1), 5)`?
- **(c)** Build a list incrementally in a loop: for i=1:5, append `i\textasciicircum{2}`

Sol 45

```r
c(list(a=1), list(b=2))                                  # (a)
```

```
# $a
# [1] 1
#
# $b
# [1] 2
```

```r
c(list(a=1), 5)                                           # (b) 5 becomes element
```

```
# $a
# [1] 1
#
# [[2]]
# [1] 5
```

```r
l <- list()
for (i in 1:5) l[[i]] <- i^2; l                          # (c)
```

```
# [[1]]
# [1] 1
#
# [[2]]
# [1] 4
#
# [[3]]
# [1] 9
```

Ex 46: Conversion Table

Problem

- **(a)** Vector to list: `as.list(1:3)` vs `list(1:3)`. Explain difference
- **(b)** List to data frame: `as.data.frame(list(a=1:3, b=4:6))`
- **(c)** Data frame to matrix: what happens with `as.matrix(data.frame(x=1:3, y=c("a","b","c")))`?
- **(d)** List to vector: why use `unlist` not `as.vector`?

Sol 46

```r
as.list(1:3)    # 3 elements                        # (a)
```

```
# [[1]]
# [1] 1
#
# [[2]]
# [1] 2
#
# [[3]]
# [1] 3
```

```r
list(1:3)       # 1 element (the vector)
```

```
# [[1]]
# [1] 1 2 3
```

```r
as.data.frame(list(a=1:3, b=4:6))                   # (b)
```

```
#   a b
# 1 1 4
# 2 2 5
# 3 3 6
```

```r
as.matrix(data.frame(x=1:3, y=c("a","b","c")))    # (c) all char!
```

```
#      x   y
# [1,] "1" "a"
```

Ex 47: Named List as Lookup Table

Problem

- **(a)** Create a lookup: `grade\_points <- list(A=4.0, B=3.0, C=2.0, D=1.0, F=0)`
- **(b)** Look up GPA for grades `c("A","B","F","A","C")` using `sapply` + `[[`
- **(c)** Compute mean GPA

Sol 47

```r
grade_points <- list(A=4.0, B=3.0, C=2.0, D=1.0, F=0)
g <- c("A","B","F","A","C")
gpa <- sapply(g, function(x) grade_points[[x]])    # (b)
gpa
```

```
# A B F A C
# 4 3 0 4 2
```

```r
mean(gpa)                                          # (c) 2.6
```

```
# [1] 2.6
```

Ex 48: `lapply/sapply` on Lists

Problem

`l <- list(a=1:10, b=rnorm(20), c=c(TRUE,FALSE,TRUE,TRUE)).`

- **(a)** Compute `mean` of each element with `lapply` and `sapply`
- **(b)** Compute `length` of each
- **(c)** Apply custom function: range (max - min)

Sol 48

```r
set.seed(1)
l <- list(a=1:10, b=rnorm(20), c=c(TRUE,FALSE,TRUE,TRUE))
lapply(l, mean); sapply(l, mean)                    # (a)
```

```
# $a
# [1] 5.5
#
# $b
# [1] 0.1905239
#
# $c
# [1] 0.75

#         a         b         c
# 5.5000000 0.1905239 0.7500000
```

```r
sapply(l, length)                                   # (b)
```

```
#  a  b  c
# 10 20  4
```

```r
sapply(l, function(x) diff(range(x)))               # (c)
```

```
#        a        b        c
# 9.000000 3.809981 1.000000
```

Ex 49: `Filter`, `Find`, `Position`

Problem

`l <- list(1:3, "hello", 4:6, TRUE, NULL, c(1.5, 2.5))`.

- **(a)** Keep only numeric elements
- **(b)** Find first character element
- **(c)** Position of first character element
- **(d)** Filter integers from `1:30` that are divisible by both 3 and 7

Sol 49

```r
l <- list(1:3, "hello", 4:6, TRUE, NULL, c(1.5, 2.5))
Filter(is.numeric, l)                                # (a)
```

```
# [[1]]
# [1] 1 2 3
#
# [[2]]
# [1] 4 5 6
#
# [[3]]
# [1] 1.5 2.5
```

```r
Find(is.character, l)                                # (b) "hello"
```

```
# [1] "hello"
```

```r
Position(is.character, l)                            # (c) 2
```

```
# [1] 2
```

```r
Filter(\(x) x%%3==0 & x%%7==0, 1:30)                # (d) 21
```

```
# [1] 21
```

Ex 50: `Reduce()`

Problem

- **(a)** Compute `1+2+3+4+5` using `Reduce("+", 1:5)`. Show accumulated steps
- **(b)** Find intersection of 3 vectors via `Reduce(intersect, ...)`
- **(c)** Merge 3 data frames pairwise using `Reduce`

Sol 50

```r
Reduce("+", 1:5)                                    # (a) 15
```

```
# [1] 15
```

```r
Reduce("+", 1:5, accumulate=TRUE)                   # steps
```

```
# [1]  1  3  6 10 15
```

```r
vecs <- list(c(1,2,3,4), c(2,3,4,5), c(3,4,5,6))
Reduce(intersect, vecs)                             # (b) 3 4
```

```
# [1] 3 4
```

```r
dfs <- list(data.frame(id=1:3, a=10:12),
            data.frame(id=2:4, b=20:22))
Reduce(\(x,y) merge(x,y,by="id",all=TRUE), dfs)   # (c)
```

```
#   id  a  b
# 1  1 10 NA
# 2  2 11 20
# 3  3 12 21
# 4  4 NA 22
```

Ex 51: List Modification Patterns

Problem

- **(a)** Given `l <- list(x=1, y=2, z=3)`, remove `x` and `z` in one operation
- **(b)** Replace all numeric elements with their squares using `lapply`
- **(c)** Deep copy a nested list and modify without affecting original

Sol 51

```r
l <- list(x=1, y=2, z=3)
l[c("x","z")] <- NULL; l                          # (a) only y
```

```
# $y
# [1] 2
```

```r
l2 <- list(a=3, b="hi", c=4)
lapply(l2, function(x) if(is.numeric(x)) x^2 else x) # (b)
```

```
# $a
# [1] 9
#
# $b
# [1] "hi"
#
# $c
# [1] 16
```

```r
orig <- list(inner=list(val=10))                  # (c)
copy <- orig; copy$inner$val <- 99
orig$inner$val  # still 10 (R copies on modify)
```

```
# [1] 10
```

Ex 52: `rapply()` — Recursive Apply

Problem

`l <- list(a=1:3, b=list(c=4:6, d=list(e=7:9)))`.

- **(a)** Apply `sum` to all leaf elements using `rapply(l, sum)`
- **(b)** Use `rapply(l, sum, how="unlist")` vs `how="list"`
- **(c)** Double all values with `rapply(l, function(x) x*2, how="replace")`

Sol 52

```r
l <- list(a=1:3, b=list(c=4:6, d=list(e=7:9)))
rapply(l, sum)                                    # (a)
```

```
#     a   b.c b.d.e
#     6    15    24
```

```r
rapply(l, sum, how="unlist")                      # (b) named vector
```

```
#     a   b.c b.d.e
#     6    15    24
```

```r
rapply(l, sum, how="list")                        # list structure
```

```
# $a
# [1] 6
#
# $b
# $b$c
# [1] 15
#
# $b$d
# $b$d$e
# [1] 24
```

```r
str(rapply(l, \(x) x*2, how="replace"))           # (c) doubled
```

```
# List of 2
```

Matrices (Ex 53–58)

Ex 53: Creating Matrices

Problem

- **(a)** Create 3×4 matrix filled column-wise from `1:12`
- **(b)** Create same matrix filled row-wise
- **(c)** Name rows `r1,r2,r3` and columns `c1,...,c4`. Access `m["r2","c3"]`
- **(d)** Create a 3×3 identity matrix with `diag`

Sol 53

```r
matrix(1:12, nrow=3)                                    # (a)
```

```
#      [,1] [,2] [,3] [,4]
# [1,]    1    4    7   10
# [2,]    2    5    8   11
# [3,]    3    6    9   12
```

```r
matrix(1:12, nrow=3, byrow=TRUE)                        # (b)
```

```
#      [,1] [,2] [,3] [,4]
# [1,]    1    2    3    4
# [2,]    5    6    7    8
# [3,]    9   10   11   12
```

```r
m <- matrix(1:12, 3, 4)
rownames(m) <- paste0("r",1:3); colnames(m) <- paste0("c",1:4)
m["r2","c3"]                                            # (c) 8
```

```
# [1] 8
```

```r
diag(3)                                                 # (d)
```

```
#      [,1] [,2] [,3]
# [1,]    1    0    0
# [2,]    0    1    0
# [3,]    0    0    1
```

Ex 54: Matrix Arithmetic

Problem

`A <- matrix(1:4, 2, 2); B <- matrix(5:8, 2, 2).`

- **(a)** Element-wise `*` vs matrix `%*%`
- **(b)** Compute `t(A)`, `det(A)`, `solve(A)`
- **(c)** Verify `A %*% solve(A)` is the identity
- **(d)** `crossprod(A,B)` vs `t(A) %*% B`

Sol 54

```r
A <- matrix(1:4, 2, 2); B <- matrix(5:8, 2, 2)
A * B; A %*% B                                          # (a)
```

```
#      [,1] [,2]
# [1,]    5   21
# [2,]   12   32

#      [,1] [,2]
# [1,]   23   31
# [2,]   34   46
```

```r
t(A); det(A); solve(A)                                  # (b)
```

```
#      [,1] [,2]
# [1,]    1    2
# [2,]    3    4

# [1] -2

#      [,1] [,2]
# [1,]   -2  1.5
# [2,]    1 -0.5
```

```r
round(A %*% solve(A))                                    # (c) identity
```

```
#      [,1] [,2]
# [1,]    1    0
```

Ex 55: `apply()` on Matrices

Problem

`m <- matrix(1:12, nrow=3)`.

- **(a)** Row sums and column sums via `apply`
- **(b)** Per-column range
- **(c)** Standardize each column (subtract mean, divide by sd)
- **(d)** Compare `apply(m,2,sum)` speed with `colSums(m)`

Sol 55

```r
m <- matrix(1:12, nrow=3)
apply(m, 1, sum); apply(m, 2, sum)                    # (a)
```

```
# [1] 22 26 30

# [1]  6 15 24 33
```

```r
apply(m, 2, range)                                     # (b)
```

```
#      [,1] [,2] [,3] [,4]
# [1,]    1    4    7   10
# [2,]    3    6    9   12
```

```r
apply(m, 2, \(x) (x-mean(x))/sd(x))                   # (c)
```

```
#      [,1] [,2] [,3] [,4]
# [1,]   -1   -1   -1   -1
# [2,]    0    0    0    0
# [3,]    1    1    1    1
```

```r
all.equal(apply(m,2,sum), colSums(m))                  # (d) TRUE
```

```
# [1] TRUE
```

Ex 56: `outer()` Product

Problem

- **(a)** Multiplication table 1..5 $\times$ 1..5 using `outer`
- **(b)** Compute $f(x, y) = x^2 + y^2$ for `x=1:4`, `y=1:3`
- **(c)** All pairwise `paste` of `c("a","b")` and `c("1","2","3")`

Sol 56

```r
outer(1:5, 1:5, "*")                                  # (a)
```

```
#      [,1] [,2] [,3] [,4] [,5]
# [1,]    1    2    3    4    5
# [2,]    2    4    6    8   10
# [3,]    3    6    9   12   15
# [4,]    4    8   12   16   20
# [5,]    5   10   15   20   25
```

```r
outer(1:4, 1:3, \(x,y) x^2 + y^2)                    # (b)
```

```
#      [,1] [,2] [,3]
# [1,]    2    5   10
# [2,]    5    8   13
# [3,]   10   13   18
# [4,]   17   20   25
```

```r
outer(c("a","b"), c("1","2","3"), paste0)             # (c)
```

```
#      [,1] [,2] [,3]
# [1,] "a1" "a2" "a3"
# [2,] "b1" "b2" "b3"
```

Ex 57: Matrix Row/Column Selection

Problem

`m <- matrix(1:12, 3, 4)`.

- **(a)** Extract row 2 as a vector vs as a 1-row matrix (`drop`)
- **(b)** Extract column 3 both ways
- **(c)** Select submatrix: rows 1-2, columns 2-3
- **(d)** Replace all elements > 8 with `NA`

Sol 57

```r
m <- matrix(1:12, 3, 4)
m[2,]; m[2,,drop=FALSE]                                  # (a)
```

```
# [1]  2  5  8 11

#      [,1] [,2] [,3] [,4]
# [1,]    2    5    8   11
```

```r
m[,3]; m[,3,drop=FALSE]                                  # (b)
```

```
# [1] 7 8 9

#      [,1]
# [1,]    7
# [2,]    8
# [3,]    9
```

```r
m[1:2, 2:3]                                              # (c)
```

```
#      [,1] [,2]
# [1,]    4    7
# [2,]    5    8
```

```r
m[m > 8] <- NA; m                                        # (d)
```

```
#      [,1] [,2] [,3] [,4]
# [1,]    1    4    7   NA
```

Ex 58: Linear System

Problem

Solve $2x + 3y = 7,\ 4x - y = 1$.

- **(a)** Set up as `A %*% x = b` and solve with `solve(A, b)`
- **(b)** Verify by multiplying back
- **(c)** Compute eigenvalues of `A`

Sol 58

```r
A <- matrix(c(2,4,3,-1), 2, 2); b <- c(7,1)
x <- solve(A, b); x                                  # (a)
```

```
# [1] 0.7142857 1.8571429
```

```r
A %*% x  # should be c(7,1)                          # (b)
```

```
#      [,1]
# [1,]    7
# [2,]    1
```

```r
eigen(A)$values                                       # (c)
```

```
# [1]  4.274917 -3.274917
```

Programming (Ex 59–84)

Ex 59: Default Arguments

Problem

- **(a)** Write `power(x, n=2)` that computes `x\textasciicircum{n}`. Test with 1 and 2 arguments
- **(b)** Write `greet(name, greeting="Hello")` that returns `"Hello, Alice!"`
- **(c)** What happens if you call `power(n=3)` without `x`?

Sol 59

```r
power <- function(x, n=2) x^n
power(3); power(3, 3); power(x=5, n=2)            # (a)
```

```
# [1] 9

# [1] 27

# [1] 25
```

```r
greet <- function(name, greeting="Hello")
  paste0(greeting, ", ", name, "!")
greet("Alice"); greet("Bob", "Hi")                 # (b)
```

```
# [1] "Hello, Alice!"

# [1] "Hi, Bob!"
```

```r
power(n=3)  # error: x is missing                  # (c)
```

```
# Error in power(n = 3): argument "x" is missing, with no default
```

Ex 60: Functions as Arguments

Problem

- **(a)** Write `apply\_fn(x, fn)` that applies `fn` to `x`. Test with `sqrt`, `log`, `abs`
- **(b)** Extend to `apply\_fn(x, fn, ...)` supporting extra arguments. Test: `apply\_fn(c(1,NA,3), mean, na.rm=TRUE)`

Sol 60

```r
apply_fn <- function(x, fn, ...) fn(x, ...)
apply_fn(c(4,9,16), sqrt)                          # (a)
```

```
# [1] 2 3 4
```

```r
apply_fn(c(1,NA,3), mean, na.rm=TRUE)              # (b) 2
```

```
# [1] 2
```

Ex 61: Implicit Return

Problem

- **(a)** Write two versions of average: one with explicit `return()`, one without
- **(b)** Write `bad\_avg` that assigns to a local variable as last line. Show it returns silently
- **(c)** Capture the invisible result with `x <- bad\_avg(1:10)`

Sol 61

```r
avg_explicit <- function(x) { return(sum(x)/length(x)) }
avg_implicit <- function(x) { sum(x)/length(x) }
avg_explicit(1:10); avg_implicit(1:10)             # (a) same
```

```
# [1] 5.5
```

```
# [1] 5.5
```

```r
bad_avg <- function(x) { result <- sum(x)/length(x) }
bad_avg(1:10)  # no output!                        # (b)
x <- bad_avg(1:10); x  # but value is captured     # (c)
```

```
# [1] 5.5
```

Ex 62: Anonymous Functions

Problem

- **(a)** Use `sapply(1:5, function(x) x\textasciicircum{2+1})`
- **(b)** Rewrite with `\textbackslash\{\(x)}` shorthand
- **(c)** Use anonymous function in `Filter` to keep even numbers from `1:20`
- **(d)** Nested: `sapply(1:3, \textbackslash\{\(i) sapply(1:3, \{}(j) i*j))}`

Sol 62

```r
sapply(1:5, function(x) x^2+1)                    # (a)
```

```
# [1]  2  5 10 17 26
```

```r
sapply(1:5, \(x) x^2+1)                           # (b)
```

```
# [1]  2  5 10 17 26
```

```r
Filter(\(x) x%%2==0, 1:20)                        # (c)
```

```
#  [1]  2  4  6  8 10 12 14 16 18 20
```

```r
sapply(1:3, \(i) sapply(1:3, \(j) i*j))           # (d) 3x3 matrix
```

```
#      [,1] [,2] [,3]
# [1,]    1    2    3
# [2,]    2    4    6
# [3,]    3    6    9
```

Ex 63: Closures

Problem

- **(a)** Write `make\_adder(n)` that returns a function adding `n`. Test `add5 <- make\_adder(5); add5(3)`
- **(b)** Write `make\_counter()` returning a list with `increment`, `get`, `reset` functions using `<<-`
- **(c)** Verify the counter works across multiple calls

Sol 63

```r
make_adder <- function(n) function(x) x + n        # (a)
add5 <- make_adder(5); add5(3)                      # 8
```

```
# [1] 8
```

```r
make_counter <- function() {                        # (b)
  n <- 0
  list(inc = function() {n <<- n+1; n},
       get = function() n,
       reset = function() {n <<- 0})
}
ctr <- make_counter()
ctr$inc(); ctr$inc(); ctr$get()                     # (c) 2
```

```
# [1] 1

# [1] 2

# [1] 2
```

```r
ctr$reset(); ctr$get()                              # 0
```

```
# [1] 0
```

Ex 64: Dot-Prefix Naming

Problem

- **(a)** Define `.helper <- function(x) x+1` and `public <- function(x) .helper(x)*2`
- **(b)** Show `.helper` doesn't appear in `ls()` but does in `ls(all.names=TRUE)`
- **(c)** Name 3 built-in R objects starting with `.`

Sol 64

```r
e <- new.env()
e$.helper <- function(x) x+1
e$public  <- function(x) e$.helper(x)*2
e$public(5)  # 12                                  # (a)
```

```
# [1] 12
```

```r
ls(e); ls(e, all.names=TRUE)                        # (b)
```

```
# [1] "public"

# [1] ".helper" "public"
```

```r
# (c) .Machine, .Platform, .GlobalEnv
```

Ex 65: `if (FALSE)` Idiom

Problem

- **(a)** Wrap code in `if (FALSE) \{ ... \}` and show it doesn't execute
- **(b)** Change to `if (TRUE)` and show it runs
- **(c)** Why is this better than commenting with `\#` for multi-line blocks?

Sol 65

```r
if (FALSE) {                                        # (a)
  cat("This won't print\n")
  x <- 999
}
exists("x")  # from previous chunks, may be TRUE
```

```
# [1] TRUE
```

```r
if (TRUE) { cat("This prints\n") }                  # (b)
```

```
# This prints
```

```r
# (c) Unlike #, the code inside is still parsed by R,
# so syntax errors are caught immediately.
```

Ex 66: Local Helper Functions

Problem

- **(a)** Write `outer\_fn(x)` that defines inner helper `validate(v)` checking `is.numeric(v) \&\& length(v) > 0`, then uses it
- **(b)** Show that `validate` is not accessible outside `outer\_fn`
- **(c)** Why is this pattern useful?

Sol 66

```r
outer_fn <- function(x) {
  validate <- function(v) is.numeric(v) && length(v) > 0
  if (!validate(x)) stop("invalid input")
  mean(x)
}
outer_fn(1:10)                                          # (a) 5.5
```

```
# [1] 5.5
```

```r
outer_fn("abc")                                         # error
```

```
# Error in outer_fn("abc"): invalid input
```

```r
# (b) validate is local, not in global env
# (c) keeps global namespace clean; helper has access to
#     outer function's variables via lexical scoping
```

Ex 67: `if/else` and `ifelse`

Problem

- **(a)** Write scalar `classify(x)` returning "pos", "neg", or "zero" using `if/else`
- **(b)** Vectorize: classify `c(-3,0,5,2,-1)` using `ifelse`
- **(c)** Nested `ifelse` to map scores to grades: ≥ 90 A, ≥ 80 B, ≥ 70 C, else F

Sol 67

```r
classify <- function(x) {                          # (a)
  if (x > 0) "pos" else if (x < 0) "neg" else "zero"
}
classify(5); classify(-1); classify(0)
```

```
# [1] "pos"

# [1] "neg"

# [1] "zero"
```

```r
ifelse(c(-3,0,5,2,-1)>0, "pos",
  ifelse(c(-3,0,5,2,-1)<0, "neg", "zero"))         # (b)
```

```
# [1] "neg"  "zero" "pos"  "pos"  "neg"
```

```r
s <- c(95,82,71,65,90)                             # (c)
ifelse(s>=90,"A",ifelse(s>=80,"B",ifelse(s>=70,"C","F")))
```

```
# [1] "A" "B" "C" "F" "A"
```

Ex 68: `switch()`

Problem

- **(a)** Write `calc(op, a, b)` using `switch(op, "+"=a+b, "-"=a-b, "*"=a*b, "/"=a/b, stop("unknown"))`
- **(b)** Test with all 4 operators and an invalid one
- **(c)** Write `day\_type(day)` classifying weekdays vs weekends

Sol 68

```r
calc <- function(op, a, b)                          # (a)
  switch(op, "+"=a+b, "-"=a-b, "*"=a*b, "/"=a/b,
         stop("unknown op"))
calc("+",3,4); calc("*",5,6); calc("/",10,3)      # (b)
```

```
# [1] 7

# [1] 30

# [1] 3.333333
```

```r
calc("^",2,3)  # error
```

```
# Error in calc("^", 2, 3): unknown op
```

```r
day_type <- function(d) switch(d, Monday=,Tuesday=,# (c)
  Wednesday=,Thursday=,Friday="Weekday",
  Saturday=,Sunday="Weekend","Unknown")
day_type("Monday"); day_type("Sunday")
```

```
# [1] "Weekday"

# [1] "Weekend"
```

Ex 69: && / || vs & / |

Problem

- **(a)** Show `\&\&` short-circuits: `FALSE \&\& stop("boom")` doesn't error
- **(b)** Show `\&` doesn't short-circuit: `c(FALSE,TRUE) \& c(TRUE,TRUE)` is vectorized
- **(c)** Write safe null-check: `if (!is.null(x) \&\& x > 0)` and explain why `\&\&` is required

Sol 69

```r
FALSE && stop("boom")  # no error: short-circuit   # (a)
```

```
# [1] FALSE
```

```r
c(FALSE,TRUE) & c(TRUE,TRUE)                       # (b) vectorized
```

```
# [1] FALSE  TRUE
```

```r
x <- NULL
if (!is.null(x) && x > 0) cat("pos") else cat("safe") # (c)
```

```
# safe
```

```r
# && stops after !is.null(x) is FALSE; & would try x>0
```

Ex 70: `for` Loop — FizzBuzz

Problem

- **(a)** Classic FizzBuzz 1-20: print "Fizz" if divisible by 3, "Buzz" if by 5, "FizzBuzz" if both, else the number
- **(b)** Vectorized version using `ifelse` (no loop)
- **(c)** Compare with `paste0` trick

Sol 70

```r
for (i in 1:20) {                                    # (a)
  if (i%%15==0) cat("FizzBuzz ")
  else if (i%%3==0) cat("Fizz ")
  else if (i%%5==0) cat("Buzz ")
  else cat(i, "")
}
```

```
# 1 2 Fizz 4 Buzz Fizz 7 8 Fizz Buzz 11 Fizz 13 14 FizzBuzz 16 17 Fizz 19 Buzz
```

```r
cat("\n")
```

```r
x <- 1:20                                            # (b)
ifelse(x%%15==0,"FizzBuzz",ifelse(x%%3==0,"Fizz",
  ifelse(x%%5==0,"Buzz",as.character(x))))
```

```
#  [1] "1"        "2"        "Fizz"     "4"        "Buzz"     "Fizz"
#  [7] "7"        "8"        "Fizz"     "Buzz"     "11"       "Fizz"
# [13] "13"       "14"       "FizzBuzz" "16"       "17"       "Fizz"
# [19] "19"       "Buzz"
```

Ex 71: `while` Loop — Newton's Method

Problem

Implement Newton's method for $\sqrt{a}$: $x_{n+1} = \frac{1}{2}(x_n + a/x_n)$.

- **(a)** Write `my\_sqrt(a, tol=1e-10)` using `while`
- **(b)** Test on 2, 9, 100
- **(c)** Count iterations needed for a=2

Sol 71

```r
my_sqrt <- function(a, tol=1e-10) {
  x <- a; n <- 0
  while (TRUE) {
    x_new <- 0.5 * (x + a/x); n <- n + 1
    if (abs(x_new - x) < tol) break
    x <- x_new
  }
  list(root=x_new, iterations=n)
}
my_sqrt(2); my_sqrt(9)$root; my_sqrt(100)$root     # (a)(b)
```

```
# $root
# [1] 1.414214
#
# $iterations
# [1] 5

# [1] 3

# [1] 10
```

```r
my_sqrt(2)$iterations                              # (c)
```

```
# [1] 5
```

Ex 72: `repeat/break` — Collatz

Problem

Collatz: if n even, $n/2$; if odd, $3n + 1$. Conjecture: always reaches 1.

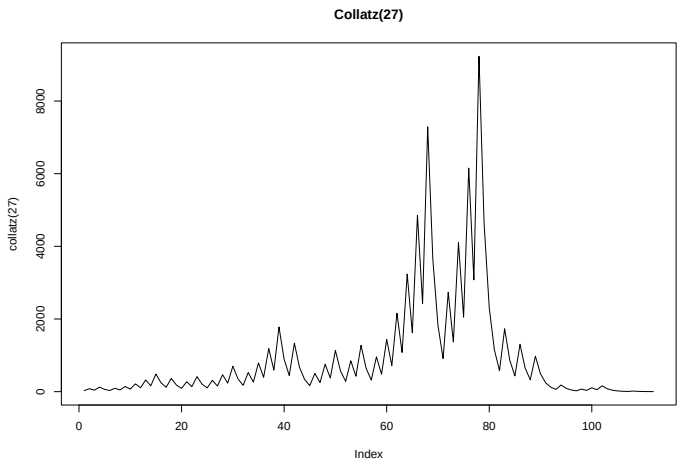
- **(a)** Write `collatz(n)` returning the sequence using `repeat/break`
- **(b)** Find the starting number 1–100 with the longest sequence
- **(c)** Plot the sequence for `n=27`

Sol 72

```
collatz <- function(n) {
  seq <- n
  repeat { if (n==1) break; n <- if(n%%2==0) n/2 else 3*n+1
           seq <- c(seq, n) }
  seq
}
lengths <- sapply(1:100, \(n) length(collatz(n)))  # (b)
which.max(lengths)  # starting number
```

```
# [1] 97
```

```
plot(collatz(27), type="l", main="Collatz(27)")    # (c)
```



Ex 73: `next` — Filtered Sum

Problem

- **(a)** Sum integers 1-100 that are NOT multiples of 7, using `for+next`
- **(b)** Same with vectorized filtering (no loop)
- **(c)** Sum only prime numbers 1-50. Write `is\_prime(n)` helper

Sol 73

```r
s <- 0; for(i in 1:100) {if(i%%7==0) next; s<-s+i}; s # (a)
```

```
# [1] 4315
```

```r
sum((1:100)[(1:100)%%7 != 0])                          # (b) same
```

```
# [1] 4315
```

```r
is_prime <- function(n) {
  if (n < 2) return(FALSE)
  if (n == 2) return(TRUE)
  all(n %% 2:floor(sqrt(n)) != 0)
}
sum(Filter(is_prime, 1:50))                            # (c)
```

```
# [1] 325
```

Ex 74: Vectorization Speed

Problem

- **(a)** Write loop version of `sum(x\textasciicircum{2)}` for `x <- 1:1e5`
- **(b)** Compare `system.time` with vectorized `sum(x\textasciicircum{2)}`
- **(c)** Pre-allocate vs grow-in-loop: time `v <- c(); for(...) v <- c(v,i)` vs `v <- numeric(n); for(...) v[i] <- i`

Sol 74

```r
n <- 1e5; x <- 1:n
loop_sum <- function(x) {s<-0; for(v in x) s<-s+v^2; s}
system.time(loop_sum(x))                              # (a) slow
```

```
#    user  system elapsed
#   0.004   0.001   0.004
```

```r
system.time(sum(x^2))                                 # (b) fast
```

```
#    user  system elapsed
#       0       0       0
```

```r
system.time({v<-c(); for(i in 1:n) v<-c(v,i)})    # (c) very slow
```

```
#    user  system elapsed
#   4.640   0.036   4.676
```

```r
system.time({v<-numeric(n); for(i in 1:n) v[i]<-i}) # faster
```

```
#    user  system elapsed
#   0.004   0.000   0.004
```

Ex 75: Sieve of Eratosthenes

Problem

- **(a)** Implement `sieve(n)` returning all primes up to `n` using a logical vector and nested loops
- **(b)** Test with `sieve(50)`
- **(c)** How many primes below 1000?

Sol 75

```r
sieve <- function(n) {
  is_p <- rep(TRUE, n); is_p[1] <- FALSE
  for (i in 2:floor(sqrt(n)))
    if (is_p[i]) is_p[seq(i*2, n, i)] <- FALSE
  which(is_p)
}
sieve(50)                                          # (b)
```

```
#  [1]  2  3  5  7 11 13 17 19 23 29 31 37 41 43 47
```

```r
length(sieve(1000))                                # (c) 168
```

```
# [1] 168
```

Ex 76: Recursive Factorial and Fibonacci

Problem

- **(a)** Write `my\_fact(n)` using `Recall`
- **(b)** Write `fib(n)` using `Recall`
- **(c)** Compute `sapply(1:10, fib)`
- **(d)** Why is naive recursive `fib` slow for large `n`?

Sol 76

```r
my_fact <- function(n) {                                # (a)
  if (n <= 1) return(1); n * Recall(n-1)
}
my_fact(5); my_fact(10)
```

```
# [1] 120

# [1] 3628800
```

```r
fib <- function(n) {                                    # (b)
  if (n <= 2) return(1); Recall(n-1) + Recall(n-2)
}
sapply(1:10, fib)                                       # (c)
```

```
#  [1]  1  1  2  3  5  8 13 21 34 55
```

```r
# (d) exponential time: each call branches into 2
```

Ex 77: `match.arg` and `missing`

Problem

- **(a)** Write `summary\_stat(x, type=c("mean","median","sd"))` using `match.arg`. Test partial matching
- **(b)** Write `f(x, y)` that checks `missing(y)` and uses default `y=0`
- **(c)** Show `nargs()` inside a function

Sol 77

```r
summary_stat <- function(x, type=c("mean","median","sd")) {
  type <- match.arg(type)
  switch(type, mean=mean(x), median=median(x), sd=sd(x))
}
summary_stat(1:10, "mea")                          # (a) mean
```

```
# [1] 5.5
```

```r
f <- function(x, y) {
  if (missing(y)) y <- 0; x + y
}
f(5); f(5, 3)                                      # (b)
```

```
# [1] 5

# [1] 8
```

```r
g <- function(a,b,c) nargs(); g(1); g(1,2,3)      # (c)
```

```
# [1] 1

# [1] 3
```

Ex 78: `invisible` and Side Effects

Problem

- **(a)** Write `my\_print(x)` that cats and returns `invisible(x)`. Show it doesn't print the return
- **(b)** Capture the invisible return
- **(c)** Write `log\_and\_return(x)` that prints a message and returns `invisible(NULL)`

Sol 78

```r
my_print <- function(x) { cat("Value:", x, "\n"); invisible(x) }
my_print(42)          # prints message, no [1] 42   # (a)
```

```
# Value: 42
```

```r
val <- my_print(42); val                            # (b) captured
```

```
# Value: 42

# [1] 42
```

```r
log_return <- function(x) {
  cat("Processed:", x, "\n"); invisible(NULL)
}
r <- log_return(5); is.null(r)                      # (c) TRUE
```

```
# Processed: 5

# [1] TRUE
```

Ex 79: `on.exit` with `add=TRUE`

Problem

- **(a)** Write function that sets `options(digits=3)`, uses `on.exit` to restore
- **(b)** Write function with two `on.exit(add=TRUE)` calls. Show both run
- **(c)** Show cleanup runs even on error

Sol 79

```r
show_digits <- function() {
  old <- options(digits=3)
  on.exit(options(old))
  cat("Inside:", getOption("digits"), "\n")
}
show_digits(); cat("Outside:", getOption("digits"), "\n") # (a)
```

```
# Inside: 3

# Outside: 7
```

```r
two_exits <- function() {                          # (b)
  on.exit(cat("exit1\n"), add=TRUE)
  on.exit(cat("exit2\n"), add=TRUE)
  cat("body\n")
}
two_exits()
```

```
# body
# exit1
# exit2
```

```r
tryCatch({                                          # (c)
  f <- function() { on.exit(cat("cleanup!\n")); stop("err") }
  f()
}, error = function(e) cat("caught:", e$message, "\n"))
```

Ex 80: `Vectorize()`

Problem

- **(a)** Write scalar `classify(x)` using `if/else` (not vectorized)
- **(b)** Use `Vectorize` to create vector version. Test on `c(-2, 0, 5)`
- **(c)** Why is `Vectorize` just a wrapper around `mapply`?

Sol 80

```r
classify <- function(x) {
  if (x > 0) "pos" else if (x < 0) "neg" else "zero"
}
classify_v <- Vectorize(classify)                    # (b)
classify_v(c(-2, 0, 5))
```

```
# [1] "neg"  "zero" "pos"
```

```r
# (c) Vectorize calls mapply internally, applying the
# scalar function element-wise. Not truly vectorized (still loops).
```

Ex 81: Native Pipe |>

Problem

- **(a)** Rewrite `round(mean(abs(c(-3,4,-1,2))), 2)` using `|>`
- **(b)** Pipe `mtcars` through `subset` then `head`
- **(c)** Use _ placeholder: `mtcars |> lm(mpg \textasciitilde\{\ wt, data = _)}`

Sol 81

```r
c(-3,4,-1,2) |> abs() |> mean() |> round(2)        # (a)
```

```
# [1] 2.5
```

```r
mtcars |> subset(cyl == 4) |> head(3)              # (b)
```

```
#             mpg cyl  disp hp drat   wt  qsec vs am gear carb
# Datsun 710 22.8   4 108.0 93 3.85 2.32 18.61  1  1    4    1
# Merc 240D  24.4   4 146.7 62 3.69 3.19 20.00  1  0    4    2
# Merc 230   22.8   4 140.8 95 3.92 3.15 22.90  1  0    4    2
```

```r
mtcars |> lm(mpg ~ wt, data = _) |>                # (c)
  coef()
```

```
# (Intercept)          wt
#   37.285126   -5.344472
```

Ex 82: `do.call`

Problem

- **(a)** Use `do.call(paste, c(as.list(c("a","b","c")), sep="-"))`
- **(b)** Use `do.call(rbind, list\_of\_dfs)` to stack 3 data frames
- **(c)** Build a call to `seq` dynamically from a list of arguments

Sol 82

```r
do.call(paste, c(as.list(c("a","b","c")), sep="-")) # (a)
```

```
# [1] "a-b-c"
```

```r
dfs <- list(data.frame(x=1,y=2), data.frame(x=3,y=4))
do.call(rbind, dfs)                                  # (b)
```

```
#   x y
# 1 1 2
# 2 3 4
```

```r
do.call(seq, list(from=1, to=10, by=2))            # (c)
```

```
# [1] 1 3 5 7 9
```

Ex 83: `tryCatch` and `try`

Problem

- **(a)** Use `tryCatch(log("a"), error=function(e) NA)`
- **(b)** Loop over `list(1,"a",4)`, wrapping each in `try`. Keep only successes
- **(c)** Write `safe\_div(a,b)` that stops on `b==0` and test with `tryCatch`

Sol 83

```r
tryCatch(log("a"), error=function(e) NA)            # (a)
```

```
# [1] NA
```

```r
inputs <- list(1, "a", 4)                           # (b)
results <- lapply(inputs, \(x) try(sqrt(x), silent=TRUE))
Filter(\(r) !inherits(r,"try-error"), results)
```

```
# [[1]]
# [1] 1
#
# [[2]]
# [1] 2
```

```r
safe_div <- function(a,b) {if(b==0) stop("div0!"); a/b}
tryCatch(safe_div(10,0), error=function(e) Inf)     # (c)
```

```
# [1] Inf
```

Ex 84: stopifnot and warning

Problem

- **(a)** Write `my\_mean(x)` with `stopifnot(is.numeric(x), length(x)>0)`
- **(b)** Write `safe\_sqrt(x)` that issues `warning` for negative input and returns `sqrt(abs(x))`
- **(c)** Use `suppressWarnings` to silence it

Sol 84

```r
my_mean <- function(x) {                          # (a)
  stopifnot(is.numeric(x), length(x) > 0)
  sum(x)/length(x)
}
my_mean(1:5); my_mean("abc")
```

```
# [1] 3

# Error in my_mean("abc"): is.numeric(x) is not TRUE
```

```r
safe_sqrt <- function(x) {                        # (b)
  if (any(x < 0)) warning("Negative values!")
  sqrt(abs(x))
}
safe_sqrt(c(4, -9, 16))
```

```
# [1] 2 3 4
```

```r
suppressWarnings(safe_sqrt(-1))                   # (c)
```

```
# [1] 1
```

Apply Family (Ex 85–96)

Ex 85: `apply` — Row and Column

Problem

`m <- matrix(c(3,1,4,1,5,9,2,6,5,3,5,8), nrow=3)`.

- **(a)** Column means via `apply`
- **(b)** Row ranges
- **(c)** Apply custom function returning a named vector per column
- **(d)** Apply on a data frame's numeric columns

Sol 85

```r
m <- matrix(c(3,1,4,1,5,9,2,6,5,3,5,8), nrow=3)
apply(m, 2, mean)                                    # (a)
```

```
# [1] 2.666667 5.000000 4.333333 5.333333
```

```r
apply(m, 1, range)                                   # (b) 2x3 matrix
```

```
#      [,1] [,2] [,3]
# [1,]    1    1    4
# [2,]    3    6    9
```

```r
apply(m, 2, \(x) c(mu=mean(x), sd=sd(x)))            # (c)
```

```
#        [,1] [,2]     [,3]     [,4]
# mu 2.666667    5 4.333333 5.333333
# sd 1.527525    4 2.081666 2.516611
```

```r
apply(mtcars[1:3, 1:4], 2, mean)                     # (d)
```

```
#        mpg        cyl       disp         hp
#  21.600000   5.333333 142.666667 104.333333
```

Ex 86: `lapply` and `sapply`

Problem

```
l <- list(a=1:10, b=seq(0,1,by=0.1), c=c(T,F,T,T,F)).
```

- **(a)** `sapply(l, length)`
- **(b)** `sapply(l, range)` — what shape?
- **(c)** `vapply(l, mean, numeric(1))`
- **(d)** Why prefer `vapply` over `sapply`?

Sol 86

```r
l <- list(a=1:10, b=seq(0,1,by=0.1), c=c(T,F,T,T,F))
sapply(l, length)                                    # (a) named vector
```

```
#  a  b  c
# 10 11  5
```

```r
sapply(l, range)                                     # (b) 2x3 matrix
```

```
#       a b c
# [1,]  1 0 0
# [2,] 10 1 1
```

```r
vapply(l, mean, numeric(1))                          # (c) named vector
```

```
#   a   b   c
# 5.5 0.5 0.6
```

```r
# (d) vapply guarantees return type; sapply may return list
# unexpectedly if elements have different lengths
```

Ex 87: `tapply`

Problem

- **(a)** Mean `mpg` by `cyl` in `mtcars`
- **(b)** Two-way: mean `mpg` by `cyl` $\times$ `am`
- **(c)** Use `tapply` on `ToothGrowth`: mean `len` by `supp` and `dose`

Sol 87

```r
tapply(mtcars$mpg, mtcars$cyl, mean)                # (a)
```

```
#        4        6        8
# 26.66364 19.74286 15.10000
```

```r
tapply(mtcars$mpg, list(mtcars$cyl, mtcars$am), mean) # (b)
```

```
#        0        1
# 4 22.900 28.07500
# 6 19.125 20.56667
# 8 15.050 15.40000
```

```r
tapply(ToothGrowth$len,
  list(ToothGrowth$supp, ToothGrowth$dose), mean)  # (c)
```

```
#      0.5     1     2
# OJ 13.23 22.70 26.06
# VC  7.98 16.77 26.14
```

Ex 88: `mapply` and `Map`

Problem

- **(a)** `mapply(function(x,y) x+y, 1:5, 10:14)`
- **(b)** `mapply(rep, 1:4, 4:1)`
- **(c)** `Map("+", 1:3, 10:12)` — what type?
- **(d)** Use `Map` to apply `paste` element-wise to two vectors

Sol 88

```r
mapply(function(x,y) x+y, 1:5, 10:14)              # (a)
```

```
# [1] 11 13 15 17 19
```

```r
mapply(rep, 1:4, 4:1)                               # (b) list
```

```
# [[1]]
# [1] 1 1 1 1
#
# [[2]]
# [1] 2 2 2
#
# [[3]]
# [1] 3 3
#
# [[4]]
# [1] 4
```

```r
Map("+", 1:3, 10:12)                                # (c) list
```

```
# [[1]]
# [1] 11
#
# [[2]]
# [1] 13
#
# [[3]]
```

Ex 89: `Reduce` Applications

Problem

- **(a)** Factorial via `Reduce("*", 1:6)`
- **(b)** Cumulative products via `accumulate=TRUE`
- **(c)** Find the common elements of 4 vectors using `Reduce(intersect, ...)`

Sol 89

```r
Reduce("*", 1:6)                                    # (a) 720
```

```
# [1] 720
```

```r
Reduce("*", 1:6, accumulate=TRUE)                   # (b)
```

```
# [1]   1   2   6  24 120 720
```

```r
Reduce(intersect, list(1:10, 3:12, 5:15, 7:20))   # (c) 7:10
```

```
# [1]  7  8  9 10
```

Ex 90: Functional Pipeline

Problem

Process `x <- c(-5, 3, -1, 8, -2, 7)`:

- **(a)** Filter positive → square → sum. Do with nested calls
- **(b)** Same with pipe `|>`
- **(c)** Same with `Reduce` composing 3 functions

Sol 90

```r
x <- c(-5, 3, -1, 8, -2, 7)
sum(x[x>0]^2)                                   # (a) nested
```

```
# [1] 122
```

```r
x[x>0] |> (\(v) v^2)() |> sum()                  # (b) pipe
```

```
# [1] 122
```

```r
fns <- list(\(v) v[v>0], \(v) v^2, sum)
Reduce(\(v,f) f(v), fns, init=x)                 # (c)
```

```
# [1] 122
```

Ex 91: `tapply` + `aggregate` Pipeline

Problem

Using `iris`:

- **(a)** Mean `Sepal.Length` by `Species` with `tapply`
- **(b)** Same with `aggregate`
- **(c)** Mean of all numeric columns by `Species` with `aggregate(. \textasciitilde\{\` Species, ...)}

Sol 91

```r
tapply(iris$Sepal.Length, iris$Species, mean)       # (a)
```

```
#     setosa versicolor  virginica
#      5.006      5.936      6.588
```

```r
aggregate(Sepal.Length ~ Species, data=iris, mean)  # (b)
```

```
#      Species Sepal.Length
# 1     setosa        5.006
# 2 versicolor        5.936
# 3  virginica        6.588
```

```r
aggregate(. ~ Species, data=iris, mean)             # (c)
```

```
#      Species Sepal.Length Sepal.Width Petal.Length Petal.Width
# 1     setosa        5.006       3.428        1.462       0.246
# 2 versicolor        5.936       2.770        4.260       1.326
# 3  virginica        6.588       2.974        5.552       2.026
```

Ex 92: Memoization via Closure

Problem

- **(a)** Write `make\_memo(fn)` that returns a memoized version storing results in a list environment
- **(b)** Memoize `fib`
- **(c)** Show second call is instant

Sol 92

```r
make_memo <- function(fn) {
  cache <- list()
  function(...) {
    key <- paste(..., sep=",")
    if (is.null(cache[[key]])) cache[[key]] <<- fn(...)
    cache[[key]]
  }
}
slow_fn <- function(x) { Sys.sleep(0.1); x^2 }
fast_fn <- make_memo(slow_fn)                          # (b)
system.time(fast_fn(5))                                # slow
```

```
#    user  system elapsed
#     0.0     0.0     0.1
```

```r
system.time(fast_fn(5))                                # (c) instant
```

```
#    user  system elapsed
#       0       0       0
```

Ex 93: `lapply` with Index

Problem

- **(a)** Given `l <- list(a=1:3, b=4:6)`, use `lapply(seq\_along(l), ...)` to create `"a: 1 2 3"`, `"b: 4 5 6"`
- **(b)** Same with `Map` and `names(l)`
- **(c)** Same with `mapply`

Sol 93

```r
l <- list(a=1:3, b=4:6)                                # (a)
lapply(seq_along(l), \(i)
  paste0(names(l)[i], ": ", paste(l[[i]], collapse=" ")))
```

```
# [[1]]
# [1] "a: 1 2 3"
#
# [[2]]
# [1] "b: 4 5 6"
```

```r
Map(\(nm, v) paste0(nm, ": ", paste(v,collapse=" ")), # (b)
    names(l), l)
```

```
# $a
# [1] "a: 1 2 3"
#
# $b
# [1] "b: 4 5 6"
```

```r
mapply(\(nm,v) paste0(nm,": ",paste(v,collapse=" ")), # (c)
       names(l), l)
```

```
#          a          b
# "a: 1 2 3" "b: 4 5 6"
```

Ex 94: Nested `lapply`

Problem

- **(a)** Generate a 3×3 multiplication table using `lapply(1:3, \textbackslash\{\(i)` sapply(1:3, \{}(j) i*j))}
- **(b)** Convert to a matrix with `do.call(rbind, ...)`
- **(c)** Compare with `outer(1:3, 1:3)`

Sol 94

```r
l <- lapply(1:3, \(i) sapply(1:3, \(j) i*j))        # (a)
do.call(rbind, l)                                   # (b)
```

```
#      [,1] [,2] [,3]
# [1,]    1    2    3
# [2,]    2    4    6
# [3,]    3    6    9
```

```r
outer(1:3, 1:3)                                     # (c) same
```

```
#      [,1] [,2] [,3]
# [1,]    1    2    3
# [2,]    2    4    6
# [3,]    3    6    9
```

Ex 95: Map for Parallel Processing

Problem

`prices <- list(c(100,102,98), c(50,55,52), c(200,210,205)). names_v <- c("AAPL", "GOOG", "MSFT").`

- **(a)** Compute return `diff(p)/p[-length(p)]` for each using `Map`
- **(b)** Compute mean return per stock

Sol 95

```r
prices <- list(c(100,102,98), c(50,55,52), c(200,210,205))
names_v <- c("AAPL","GOOG","MSFT")
returns <- Map(\(p) diff(p)/p[-length(p)], prices) # (a)
names(returns) <- names_v
sapply(returns, mean)                              # (b)
```

```
#         AAPL         GOOG         MSFT
# -0.009607843  0.022727273  0.013095238
```

Ex 96: `Filter` + `Reduce` Combo

Problem

- **(a)** From `1:100`, filter multiples of 3, then sum with `Reduce`
- **(b)** From a list of vectors, keep only those with length > 3, then find their common intersection
- **(c)** Compare with direct `sum(Filter(...))`

Sol 96

```r
Reduce("+", Filter(\(x) x%%3==0, 1:100))          # (a)
```

```
# [1] 1683
```

```r
vecs <- list(1:10, 3:8, 2:15, 5:12, 4:20)
long <- Filter(\(v) length(v)>3, vecs)
Reduce(intersect, long)                           # (b) 5:8
```

```
# [1] 5 6 7 8
```

```r
sum(Filter(\(x) x%%3==0, 1:100))                   # (c) same as (a)
```

```
# [1] 1683
```

Strings and Regex (Ex 97–110)

Ex 97: Basic String Functions

Problem

`x <- c("  Hello World  ", "R Programming", "data SCIENCE").`

- **(a)** Trim whitespace; count chars of each
- **(b)** Convert to uppercase, lowercase
- **(c)** `startsWith` / `endsWith` tests
- **(d)** Extract first 4 chars using `substr`

Sol 97

```r
x <- c("  Hello World  ","R Programming","data SCIENCE")
xt <- trimws(x); nchar(xt)                          # (a)
```

```
# [1] 11 13 12
```

```r
toupper(xt); tolower(xt)                            # (b)
```

```
# [1] "HELLO WORLD"   "R PROGRAMMING" "DATA SCIENCE"

# [1] "hello world"   "r programming" "data science"
```

```r
startsWith(xt, "Hello"); endsWith(xt, "SCIENCE")   # (c)
```

```
# [1]  TRUE FALSE FALSE

# [1] FALSE FALSE  TRUE
```

```r
substr(xt, 1, 4)                                    # (d)
```

```
# [1] "Hell" "R Pr" "data"
```

Ex 98: paste and sprintf

Problem

- **(a)** `paste("item", 1:5, sep="\_")`
- **(b)** Same with `collapse=", "`
- **(c)** `sprintf("Name: %s, Score: %.1f", "Alice", 95.678)`
- **(d)** Zero-pad: `sprintf("%03d", 1:5)`

Sol 98

```r
paste("item", 1:5, sep="_")                          # (a)
```

```
# [1] "item_1" "item_2" "item_3" "item_4" "item_5"
```

```r
paste("item", 1:5, sep="_", collapse=", ")           # (b)
```

```
# [1] "item_1, item_2, item_3, item_4, item_5"
```

```r
sprintf("Name: %s, Score: %.1f", "Alice", 95.678)  # (c)
```

```
# [1] "Name: Alice, Score: 95.7"
```

```r
sprintf("%03d", 1:5)                                 # (d)
```

```
# [1] "001" "002" "003" "004" "005"
```

Ex 99: `strsplit`

Problem

- **(a)** Split `"a.b.c.d"` by `"."` (note the escape!)
- **(b)** Split `"hello world foo"` by space
- **(c)** Split each character of `"ABCDE"`
- **(d)** Split `c("a-b", "c-d-e")` — what shape is the result?

Sol 99

```r
strsplit("a.b.c.d", "\\.")                          # (a)
```

```
# [[1]]
# [1] "a" "b" "c" "d"
```

```r
strsplit("hello world foo", " ")                    # (b)
```

```
# [[1]]
# [1] "hello" "world" "foo"
```

```r
strsplit("ABCDE", "")                               # (c)
```

```
# [[1]]
# [1] "A" "B" "C" "D" "E"
```

```r
strsplit(c("a-b","c-d-e"), "-")                     # (d) list of 2
```

```
# [[1]]
# [1] "a" "b"
#
# [[2]]
# [1] "c" "d" "e"
```

Ex 100: grep and grepl

Problem

```
w <- c("statistics","computation","algorithm","distribution","estimation").
```

- **(a)** Words containing "tion"
- **(b)** Words starting with vowel
- **(c)** Words of ≥ 12 characters
- **(d)** grepl to create logical mask

Sol 100

```r
w <- c("statistics","computation","algorithm",
       "distribution","estimation")
grep("tion", w, value=TRUE)                          # (a)
```

```
# [1] "computation"  "distribution" "estimation"
```

```r
grep("^[aeiou]", w, value=TRUE)                      # (b)
```

```
# [1] "algorithm"  "estimation"
```

```r
w[nchar(w) >= 12]                                    # (c)
```

```
# [1] "distribution"
```

```r
grepl("tion", w)                                     # (d)
```

```
# [1] FALSE  TRUE FALSE  TRUE  TRUE
```

Ex 101: sub and gsub

Problem

- **(a)** Replace first space with `"\_"` in `"hello world foo"`
- **(b)** Replace all spaces
- **(c)** Remove all digits from `"abc123def456"`
- **(d)** Compress multiple spaces to one

Sol 101

```r
sub(" ", "_", "hello world foo")                    # (a) first only
```

```
# [1] "hello_world foo"
```

```r
gsub(" ", "_", "hello world foo")                   # (b) all
```

```
# [1] "hello_world_foo"
```

```r
gsub("[0-9]", "", "abc123def456")                   # (c)
```

```
# [1] "abcdef"
```

```r
gsub(" +", " ", "too   many    spaces")             # (d)
```

```
# [1] "too many spaces"
```

Ex 102: Capture Groups

Problem

- **(a)** Swap first/last name: `sub("(\textbackslash\{\\{}w+) (\{}\{}w+)“,”\{}\{}2, \{}\{}1”, “John Smith”)}`
- **(b)** Extract domain from email: `sub(".*@", "", "user@example.com")`
- **(c)** Extract year from dates using capture group

Sol 102

```r
sub("(\\w+) (\\w+)", "\\2, \\1",
    c("John Smith","Jane Doe"))                    # (a)
```

```
# [1] "Smith, John" "Doe, Jane"
```

```r
sub(".*@", "", c("user@example.com","a@b.org"))    # (b)
```

```
# [1] "example.com" "b.org"
```

```r
sub("(\\d{4})-\\d{2}-\\d{2}", "\\1",
    c("2024-01-15","2023-12-31"))                  # (c) year
```

```
# [1] "2024" "2023"
```

Ex 103: `regexpr` and `gregexpr`

Problem

`x <- "Phone: 02-1234-5678, Fax: 04-8765-4321".`

- **(a)** Extract all phone numbers with `gregexpr` + `regmatches`
- **(b)** Extract first match only with `regexpr`
- **(c)** Extract all words with `\textbackslash\{\\{}b\{}\{}w+\{}\{}b}`

Sol 103

```r
x <- "Phone: 02-1234-5678, Fax: 04-8765-4321"
m <- gregexpr("[0-9]+-[0-9]+-[0-9]+", x)
regmatches(x, m)                                    # (a)
```

```
# [[1]]
# [1] "02-1234-5678" "04-8765-4321"
```

```r
m1 <- regexpr("[0-9]+-[0-9]+-[0-9]+", x)
regmatches(x, m1)                                   # (b)
```

```
# [1] "02-1234-5678"
```

```r
regmatches(x, gregexpr("\\b\\w+\\b", x))           # (c)
```

```
# [1] "Phone" "02"    "1234"  "5678"  "Fax"   "04"    "8765"  "4321"
```

Ex 104: Greedy vs Lazy

Problem

`x <- "<b>bold</b> and <i>italic</i>".`

- **(a)** `gregexpr("<.*>", x)` + `regmatches` — what matches? (greedy)
- **(b)** `gregexpr("<.*?>", x)` — what matches? (lazy)
- **(c)** Explain the difference

Sol 104

```r
x <- "<b>bold</b> and <i>italic</i>"
regmatches(x, gregexpr("<.*>", x))                # (a) one big match
```

```
# [[1]]
# [1] "<b>bold</b> and <i>italic</i>"
```

```r
regmatches(x, gregexpr("<.*?>", x))               # (b) 4 tags
```

```
# [[1]]
# [1] "<b>"  "</b>" "<i>"  "</i>"
```

```r
# (c) greedy: .* matches as much as possible (all the way
#     to last >); lazy: .*? matches as little as possible
```

Ex 105: Anchors and Quantifiers

Problem

`x <- c("cat","concatenate","scat","catalog")`.

- **(a)** Starts with `cat`
- **(b)** Ends with `cat`
- **(c)** Exactly `cat`
- **(d)** `c("a","ab","abb","abbb")`: match `ab+` vs `ab*` vs `ab?`

Sol 105

```r
x <- c("cat","concatenate","scat","catalog")
grep("^cat", x, value=TRUE)                          # (a)
```

```
# [1] "cat"     "catalog"
```

```r
grep("cat$", x, value=TRUE)                          # (b)
```

```
# [1] "cat"  "scat"
```

```r
grep("^cat$", x, value=TRUE)                         # (c)
```

```
# [1] "cat"
```

```r
y <- c("a","ab","abb","abbb")
grep("ab+", y, value=TRUE); grep("ab*", y, value=TRUE) # (d)
```

```
# [1] "ab"   "abb"  "abbb"
```

```
# [1] "a"    "ab"   "abb"  "abbb"
```

Ex 106: POSIX Classes

Problem

`x <- c("Hello!", "123", "abc", "A1!").`

- **(a)** Contains punctuation
- **(b)** All digits
- **(c)** Contains only alphabetic
- **(d)** All alphanumeric

Sol 106

```r
x <- c("Hello!", "123", "abc", "A1!")
grep("[[:punct:]]", x, value=TRUE)                # (a)
```

```
# [1] "Hello!" "A1!"
```

```r
grep("^[[:digit:]]+$", x, value=TRUE)             # (b)
```

```
# [1] "123"
```

```r
grep("^[[:alpha:]]+$", x, value=TRUE)             # (c)
```

```
# [1] "abc"
```

```r
grep("^[[:alnum:]]+$", x, value=TRUE)             # (d)
```

```
# [1] "123" "abc"
```

Ex 107: Lookahead/Lookbehind

Problem

- **(a)** Extract numbers preceded by `\$`: `c("\$100","200","\$300")`
- **(b)** Find files NOT ending in `.txt`: `c("a.txt","b.png","c.txt","d.jpg")`
- **(c)** Explain why `perl=TRUE` is needed

Sol 107

```r
x <- c("$100","200","$300")
regmatches(x, gregexpr("(?<=\\$)\\d+", x, perl=TRUE)) # (a)
```

```
# [[1]]
# [1] "100"
#
# [[2]]
# character(0)
#
# [[3]]
# [1] "300"
```

```r
y <- c("a.txt","b.png","c.txt","d.jpg")
grep("\\.(?!txt$)", y, value=TRUE, perl=TRUE)      # (b)
```

```
# [1] "b.png" "d.jpg"
```

```r
# (c) lookahead/lookbehind are Perl-compatible regex
# extensions not available in base (POSIX) regex
```

Ex 108: `regexec` — Structured Extraction

Problem

`dates <- c("2024-01-15 14:30", "2023-12-31 09:00")`.

- **(a)** Write pattern to capture year, month, day, hour, minute separately
- **(b)** Extract with `regexec` + `regmatches`
- **(c)** Build a data frame from the results

Sol 108

```r
dates <- c("2024-01-15 14:30", "2023-12-31 09:00")
pat <- "(\\d{4})-(\\d{2})-(\\d{2}) (\\d{2}):(\\d{2})"
m <- regexec(pat, dates)
parts <- regmatches(dates, m)                          # (a)(b)
do.call(rbind, lapply(parts, \(p)
  setNames(p[-1], c("Y","M","D","h","m"))))            # (c)
```

```
#      Y      M    D    h    m
# [1,] "2024" "01" "15" "14" "30"
# [2,] "2023" "12" "31" "09" "00"
```

Ex 109: camelCase snake_case

Problem

- **(a)** Write `camel\_to\_snake(s)` converting `"camelCase"` → `"camel\_case"`
- **(b)** Write `snake\_to\_camel(s)` converting `"snake\_case"` → `"snakeCase"`
- **(c)** Test both on `"myHTTPRequest"` / `"my\_http\_request"`

Sol 109

```r
camel_to_snake <- function(s) tolower(gsub("([A-Z])","_\\1",s))
camel_to_snake("camelCase")                          # (a)
```

```
# [1] "camel_case"
```

```r
camel_to_snake("myHTTPRequest")
```

```
# [1] "my_h_t_t_p_request"
```

```r
snake_to_camel <- function(s) {
  parts <- strsplit(s, "_")[[1]]
  paste0(parts[1], paste0(toupper(substring(parts[-1],1,1)),
         substring(parts[-1],2), collapse=""))
}
snake_to_camel("my_http_request")                    # (b)(c)
```

```
# [1] "myHttpRequest"
```

Ex 110: Clean Column Names

Problem

`cols <- c("First Name", "Last.Name", "Age (years)", "Income$")`.

- **(a)** Replace all non-alphanumeric chars with _
- **(b)** Remove leading/trailing _
- **(c)** Collapse consecutive _ to single _
- **(d)** Convert to lowercase

Sol 110

```r
cols <- c("First Name","Last.Name","Age (years)","Income$")
x <- gsub("[^a-zA-Z0-9]", "_", cols)              # (a)
x <- gsub("^_+|_+$", "", x)                       # (b)
x <- gsub("_+", "_", x)                           # (c)
tolower(x)                                         # (d)
```

```
# [1] "first_name" "last_name"  "age_years"  "income"
```

Date/Time, I/O, Error Handling, Advanced (Ex 111–120)

Ex 111: Date Arithmetic

Problem

- **(a)** Create today's date with `Sys.Date()`. Add 30 days
- **(b)** Days between `2024-01-01` and `2024-12-31`
- **(c)** Generate monthly sequence Jan–Jun 2024
- **(d)** Extract weekday name and month name from a date

Sol 111

```r
today <- Sys.Date(); today + 30                    # (a)
```

```
# [1] "2026-04-18"
```

```r
as.Date("2024-12-31") - as.Date("2024-01-01")     # (b) 365
```

```
# Time difference of 365 days
```

```r
seq(as.Date("2024-01-01"), as.Date("2024-06-01"),
    by="month")                                    # (c)
```

```
# [1] "2024-01-01" "2024-02-01" "2024-03-01" "2024-04-01" "2024-05-01"
# [6] "2024-06-01"
```

```r
d <- as.Date("2024-07-15")
weekdays(d); months(d); format(d, "%A %B %d, %Y") # (d)
```

```
# [1] " "

# [1] " "

# [1] "    15, 2024"
```

Ex 112: `POSIXct` and Timezones

Problem

- **(a)** Create `as.POSIXct("2024-01-15 14:30:00", tz="Asia/Taipei")`
- **(b)** Extract hour, minute via `POSIXlt`
- **(c)** Difference between two timestamps in hours
- **(d)** Format with `strftime`

Sol 112

```r
t1 <- as.POSIXct("2024-01-15 14:30:00", tz="Asia/Taipei") # (a)
lt <- as.POSIXlt(t1); lt$hour; lt$min             # (b)
```

```
# [1] 14

# [1] 30
```

```r
t2 <- as.POSIXct("2024-01-15 18:45:00", tz="Asia/Taipei")
difftime(t2, t1, units="hours")                    # (c)
```

```
# Time difference of 4.25 hours
```

```r
strftime(t1, "%Y/%m/%d %H:%M %Z")                 # (d)
```

```
# [1] "2024/01/15 14:30 CST"
```

Ex 113: Binary I/O

Problem

- **(a)** `charToRaw("Hello")` — what type?
- **(b)** Convert back with `rawToChar`
- **(c)** `as.integer(charToRaw("ABC"))` — what are the byte values?
- **(d)** Write raw bytes to a temp file with `writeBin` and read back

Sol 113

```r
r <- charToRaw("Hello"); r; typeof(r)              # (a) raw
```

```
# [1] 48 65 6c 6c 6f
```

```
# [1] "raw"
```

```r
rawToChar(r)                                        # (b) "Hello"
```

```
# [1] "Hello"
```

```r
as.integer(charToRaw("ABC"))                        # (c) 65 66 67
```

```
# [1] 65 66 67
```

```r
tmp <- tempfile()                                   # (d)
writeBin(charToRaw("test123"), tmp)
rawToChar(readBin(tmp, "raw", 20))
```

```
# [1] "test123"
```

Ex 114: `readline` and `interactive`

Problem

- **(a)** Write a function that uses `readline` to ask for user's name (use `eval=FALSE`)
- **(b)** Use `interactive()` to branch: interactive $\rightarrow$ readline, batch $\rightarrow$ default
- **(c)** Why does `readline` always return character?

Sol 114

```r
ask_name <- function() {
  if (interactive()) {
    name <- readline("Your name: ")
  } else {
    name <- "default_user"
  }
  cat("Hello,", name, "\n")
}
# (c) readline reads raw text from console; R has no way
# to know the intended type, so it always returns character
```

Ex 115: `system2`

Problem

- **(a)** Run `system2("echo", args=c("hello","world"))`
- **(b)** Capture output with `stdout=TRUE`
- **(c)** Run in background with `wait=FALSE`
- **(d)** Cross-platform check with `.Platform\$OS.type`

Sol 115

```r
system2("echo", args=c("hello","world"))            # (a)
out <- system2("date", stdout=TRUE); out            # (b)
system2("sleep", args="1", wait=FALSE)              # (c)
if (.Platform$OS.type == "unix") {                  # (d)
  system2("uname", args="-a", stdout=TRUE)
}
```

Ex 116: S3 Classes

Problem

- **(a)** Create S3 class `Circle`: `list(radius=r)` with `class="Circle"`
- **(b)** Write `print.Circle` and `area.Circle`
- **(c)** Create instance; dispatch `print` and `area`

Sol 116

```r
Circle <- function(r) structure(list(radius=r), class="Circle")
print.Circle <- function(x, ...)
  cat(sprintf("Circle(r=%.2f)\n", x$radius))
area <- function(x, ...) UseMethod("area")
area.Circle <- function(x, ...) pi * x$radius^2
c1 <- Circle(5); c1                                  # (a)(b)(c)
```

```
# Circle(r=5.00)
```

```r
area(c1)
```

```
# [1] 78.53982
```

Ex 117: Formula and `lm`

Problem

- **(a)** `f <- y \textasciitilde\{\ x}`; show `class`, `typeof`, `all.vars`
- **(b)** Fit `lm(dist \textasciitilde\{\ speed, data=cars)}`. Extract coefficients
- **(c)** Plot data with regression line

Sol 117

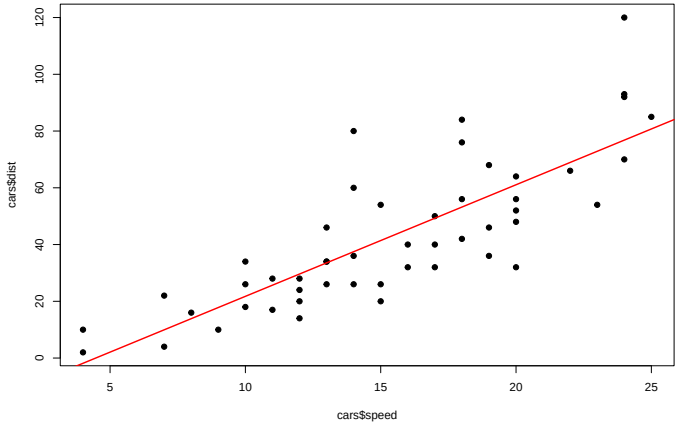
```r
f <- y ~ x; class(f); typeof(f); all.vars(f)      # (a)
```

```
# [1] "formula"

# [1] "language"

# [1] "y" "x"
```

```r
model <- lm(dist ~ speed, data=cars)
coef(model)                                       # (b)
```

```
# (Intercept)       speed
#  -17.579095    3.932409
```

```r
plot(cars$speed, cars$dist, pch=19)               # (c)
abline(model, col="red", lwd=2)
```



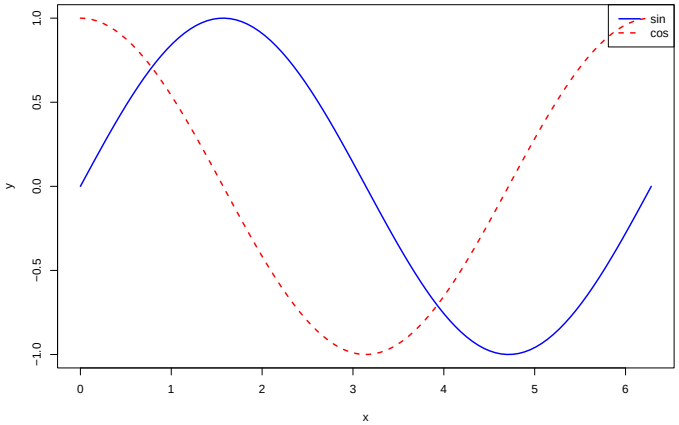
Ex 118: Plotting

Problem

- **(a)** Plot `sin(x)` and `cos(x)` for `x` from 0 to 2π with legend
- **(b)** `par(mfrow=c(1,2))`: histogram and boxplot side by side
- **(c)** Add annotation with `text()` and horizontal line with `abline(h=...)`

Sol 118

```
x <- seq(0, 2*pi, length=100)                    # (a)
plot(x, sin(x), type="l", col="blue", lwd=2, ylab="y")
lines(x, cos(x), col="red", lwd=2, lty=2)
legend("topright", c("sin","cos"), col=c("blue","red"),
       lty=1:2, lwd=2)
```



Ex 119: Random Number Generation

Problem

- **(a)** `set.seed(42)` then generate 5 uniform, 5 normal, 10 Bernoulli
- **(b)** Simulate 1000 dice rolls and tabulate
- **(c)** Estimate π via Monte Carlo: sample `(x,y)` uniform in $[0,1]^2$, count proportion inside unit circle quadrant

Sol 119

```r
set.seed(42)
runif(5); rnorm(5); rbinom(10, 1, 0.5)            # (a)
```

```
# [1] 0.9148060 0.9370754 0.2861395 0.8304476 0.6417455

# [1]  0.04788474 -1.10459944  0.53902380  0.58020632 -0.65750284

#  [1] 1 1 0 0 1 1 0 1 1 0
```

```r
table(sample(1:6, 1000, replace=TRUE))             # (b)
```

```
#
#   1   2   3   4   5   6
# 169 191 163 157 153 167
```

```r
n <- 1e5; x <- runif(n); y <- runif(n)            # (c)
4 * mean(x^2 + y^2 <= 1)  #
```

```
# [1] 3.14176
```

Ex 120: Options and Benchmarking

Problem

- **(a)** Save current `digits` option, set to 15, print `pi`, restore
- **(b)** Time `crossprod(x)` for `x <- matrix(rnorm(1e6),1000)` with `system.time`
- **(c)** Use `proc.time` for manual timing of `sum(1:1e7)`

Sol 120

```r
old <- options(digits=15); pi                          # (a)
```

```
# [1] 3.14159265358979
```

```r
options(old)
system.time({                                          # (b)
  x <- matrix(rnorm(1e6), 1000, 1000)
  y <- crossprod(x)
})
```

```
#    user  system elapsed
#   0.090   0.075   0.038
```

```r
pt <- proc.time(); s <- sum(1:1e7); proc.time()-pt # (c)
```

```
#    user  system elapsed
#   0.000   0.014   0.001
```